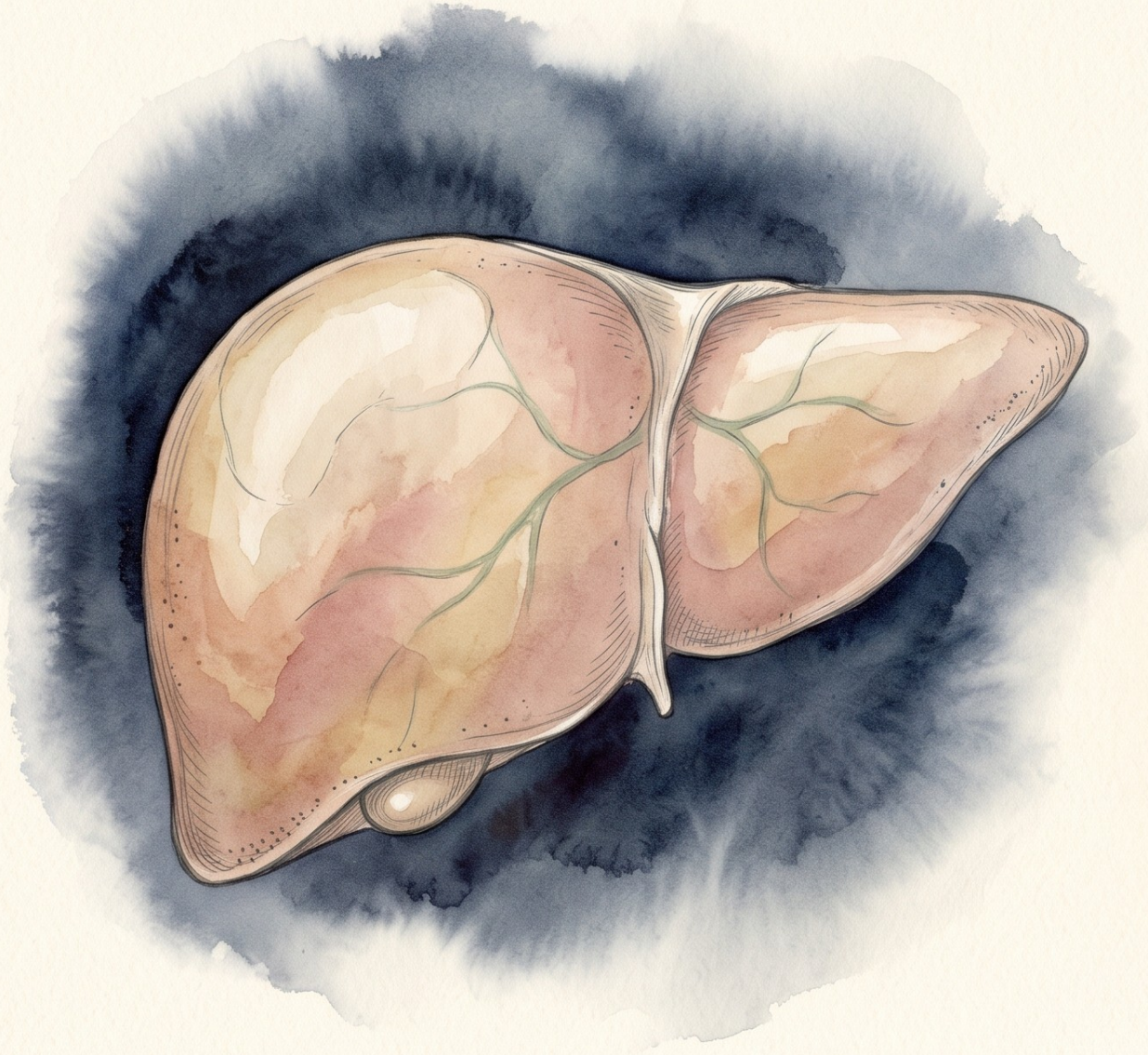


Acute Liver Failure and ACLF in the ICU

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Tufts UNIVERSITY | School of Medicine

Slides (PDF)

Illustrations generated with Google Gemini

Objectives

By the end of this talk you should be able to:

- Tell acute liver failure apart from severe acute liver injury and from ACLF
- Start the first-hour care of ALF: cause, NAC, glucose, airway
- Protect the brain in ALF and know when to intubate
- Start CRRT for ammonia without waiting for a renal indication
- Diagnose and grade ACLF, and find the trigger
- Work through AKI in cirrhosis, including HRS-AKI, without hurting the lungs
- Know when to call transplant hepatology

Two patients, same ICU

Bed 4	Bed 7
28-year-old woman	56-year-old man
No known liver disease	Alcohol-related cirrhosis with ascites
Nausea and vomiting for 4 days	Fever, confusion, low blood pressure

Both have a sick liver. Do they have the same disease?

Bed 4

28F, 4 days of nausea



28F, 4 days of nausea

- Nausea, vomiting and poor intake for 4 days
- No known liver disease, no alcohol use
- "Took something for a toothache"

Test	Result
AST	6,820 U/L
ALT	5,940 U/L
Alkaline phosphatase	128 U/L
Total bilirubin	2.9 mg/dL
Creatinine	1.0 mg/dL

Is this acute liver failure?

28F, 4 days of nausea

- Nausea, vomiting and poor intake for 4 days
- No known liver disease, no alcohol use
- "Took something for a toothache"

Test	Result
AST	6,820 U/L
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Alkaline phosphatase	128 U/L
Total bilirubin	2.9 mg/dL
Creatinine	1.0 mg/dL

Is this acute liver failure?

We don't know yet. What's missing?

28F, 4 days of nausea

- Nausea, vomiting and poor intake for 4 days
- No known liver disease, no alcohol use
- "Took something for a toothache"

Test	Result
AST	6,820 U/L
ALT	5,940 U/L
Alkaline phosphatase	128 U/L
Total bilirubin	2.9 mg/dL
Creatinine	1.0 mg/dL

What's missing?

- **INR** (synthetic function, not enzymes)
- **Mental status** (any encephalopathy?)
- **Time course** (how long since the first symptom?)
- **Is there chronic liver disease underneath?**

28F, 4 days of nausea

- Disoriented to time, asterixis. Still there after the glucose is fixed
- Symptoms started 4 days ago
- Ultrasound: smooth liver, no ascites, patent vessels, normal spleen
- No history or imaging suggesting cirrhosis

Test	Result
AST	6,820 U/L
ALT	5,940 U/L
Total bilirubin	2.9 mg/dL
INR	**3.2**
Glucose	**54 mg/dL**

28F, 4 days of nausea

- Disoriented to time, asterixis. Still there after the glucose is fixed
- Symptoms started 4 days ago
- Ultrasound: smooth liver, no ascites, patent vessels, normal spleen
- No history or imaging suggesting cirrhosis

Test	Result
AST	6,820 U/L
ALT	5,940 U/L
Total bilirubin	2.9 mg/dL
INR	**3.2**
Glucose	**54 mg/dL**

This is acute liver failure

Acute liver failure

- **INR ≥ 1.5** and **any degree of encephalopathy**
- Illness of less than 26 weeks
- No pre-existing cirrhosis
  - AASLD and ACG: chronic liver disease without cirrhosis still allows ALF. EASL: no prior liver disease at all
  - Wilson disease, autoimmune hepatitis and vertically acquired HBV can still count if recognized within 26 weeks

Jaundice to encephalopathy	**Hyperacute (< 1 week)**	**Acute (1 to 4 weeks)**	**Subacute (4 to 12 weeks)**
Typical causes	Acetaminophen, HAV, HEV, ischemia	HBV	Idiosyncratic drug injury, AIH, indeterminate
Cerebral edema	**Common**	Less common	Uncommon
Recovery without transplant	More likely	Intermediate	**Less likely**

EASL ALF guideline 2017 [PMID: 28417882]; ACG ALF 2023 [PMID: 37377263]; Management of ALF: update 2022 [PMID: 36001996]

Same patient, but wide awake

Same labs and INR 2.4, but fully oriented with no asterixis

What do you call this, and what do you do now?

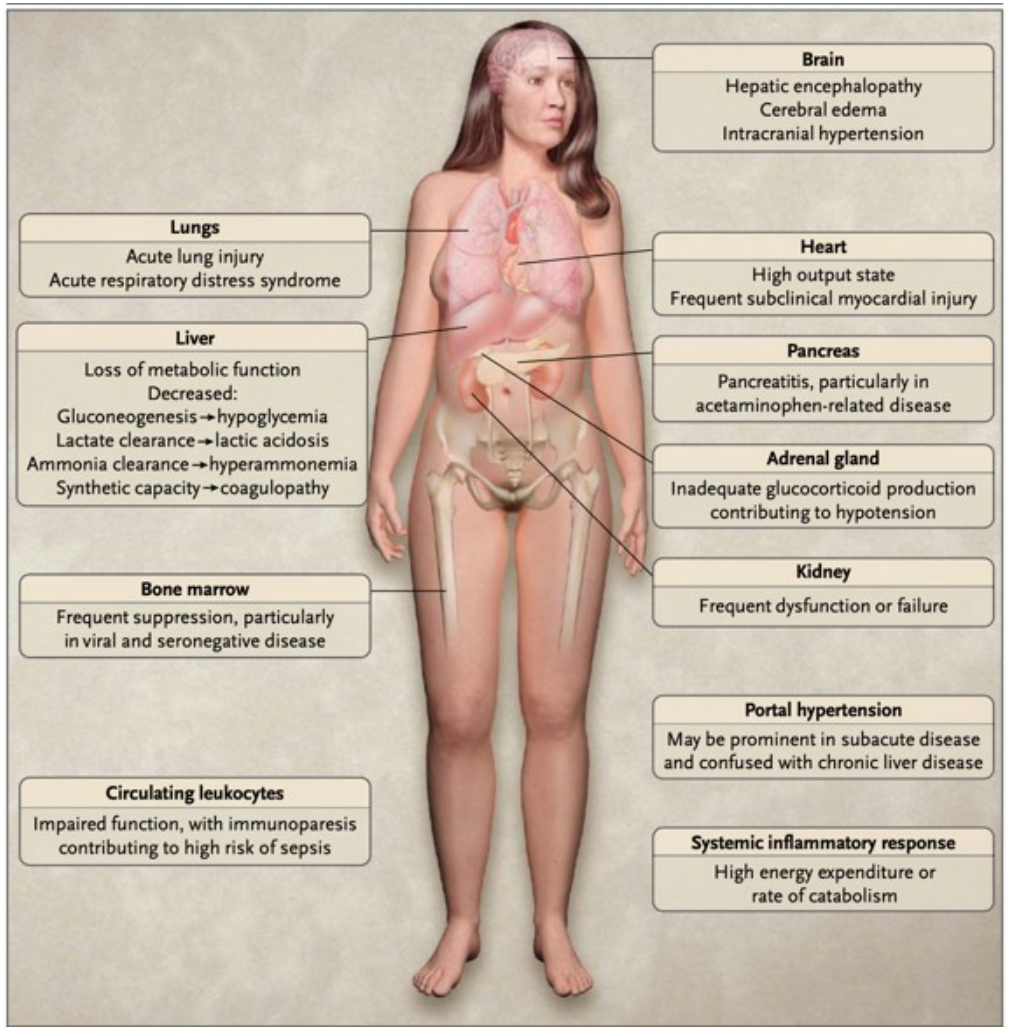
Same patient, but wide awake

Same labs and INR 2.4, but fully oriented with no asterixis

- **Severe acute liver injury**, not ALF yet
- No encephalopathy, so it doesn't meet the definition
- But it can become ALF within hours
  - Check glucose often and examine the patient every few hours
  - Call transplant hepatology early anyway
- And the opposite trap:
  - Hypoglycemia, sedatives, sepsis, alcohol withdrawal and seizures can all look like encephalopathy

Which other organs fail in ALF?

Which other organs fail in ALF?



Bernal W, Wendon J. N Engl J Med 2013;369:2525-34, Figure 1

28F, 4 days of nausea

What's your next step?

Three things, at the same time

Keep her alive	Find and treat the cause	Call transplant hepatology
ICU bed Glucose every 1 to 2 hours Airway assessment Volume, perfusion, lactate Neuro checks	Acetaminophen level Viral, autoimmune, Wilson, toxicology Doppler ultrasound Pregnancy test Start NAC if acetaminophen is possible	Now, not after the workup Transplant evaluation starts on admission Social work and psychiatry while she can still talk

All three start in the first hour

Which cause changes treatment right away?

If you suspect	Send	Don't wait for the result to
Acetaminophen	Level, ingestion history, combination products	
Herpes simplex	HSV PCR (blood)	
Hepatitis B	HBsAg, anti-HBc IgM, HBV DNA	
Autoimmune hepatitis	ANA, ASMA, IgG (can be negative)	
Wilson disease	Ceruloplasmin, urine copper, hemolysis, low ALP	
Pregnancy (AFLP, HELLP)	hCG	
Budd-Chiari	Doppler ultrasound or CT	
Ischemic hepatitis	Echo, history of shock	
Mushrooms (Amanita)	Wild mushrooms, then diarrhea 6 to 24 hours later	

Update 2022 [PMID: 36001996]; practical update 2024 [PMID: 39170946]; EASL AIH 2025 [PMID: 40348684]

Which cause changes treatment right away?

If you suspect	Send	Don't wait for the result to
Acetaminophen	Level, ingestion history, combination products	Start NAC
Herpes simplex	HSV PCR (blood)	Start IV acyclovir
Hepatitis B	HBsAg, anti-HBc IgM, HBV DNA	Start entecavir or tenofovir
Autoimmune hepatitis	ANA, ASMA, IgG (can be negative)	Discuss steroids with hepatology, list in parallel
Wilson disease	Ceruloplasmin, urine copper, hemolysis, low ALP	Prepare for urgent transplant
Pregnancy (AFLP, HELLP)	hCG	Call obstetrics, plan delivery
Budd-Chiari	Doppler ultrasound or CT	Anticoagulation, discuss TIPS or transplant
Ischemic hepatitis	Echo, history of shock	Fix the circulation
Mushrooms (Amanita)	Wild mushrooms, then diarrhea 6 to 24 hours later	NAC, call poison control (silibinin if available)

Update 2022 [PMID: 36001996]; practical update 2024 [PMID: 39170946]; EASL AIH 2025 [PMID: 40348684]

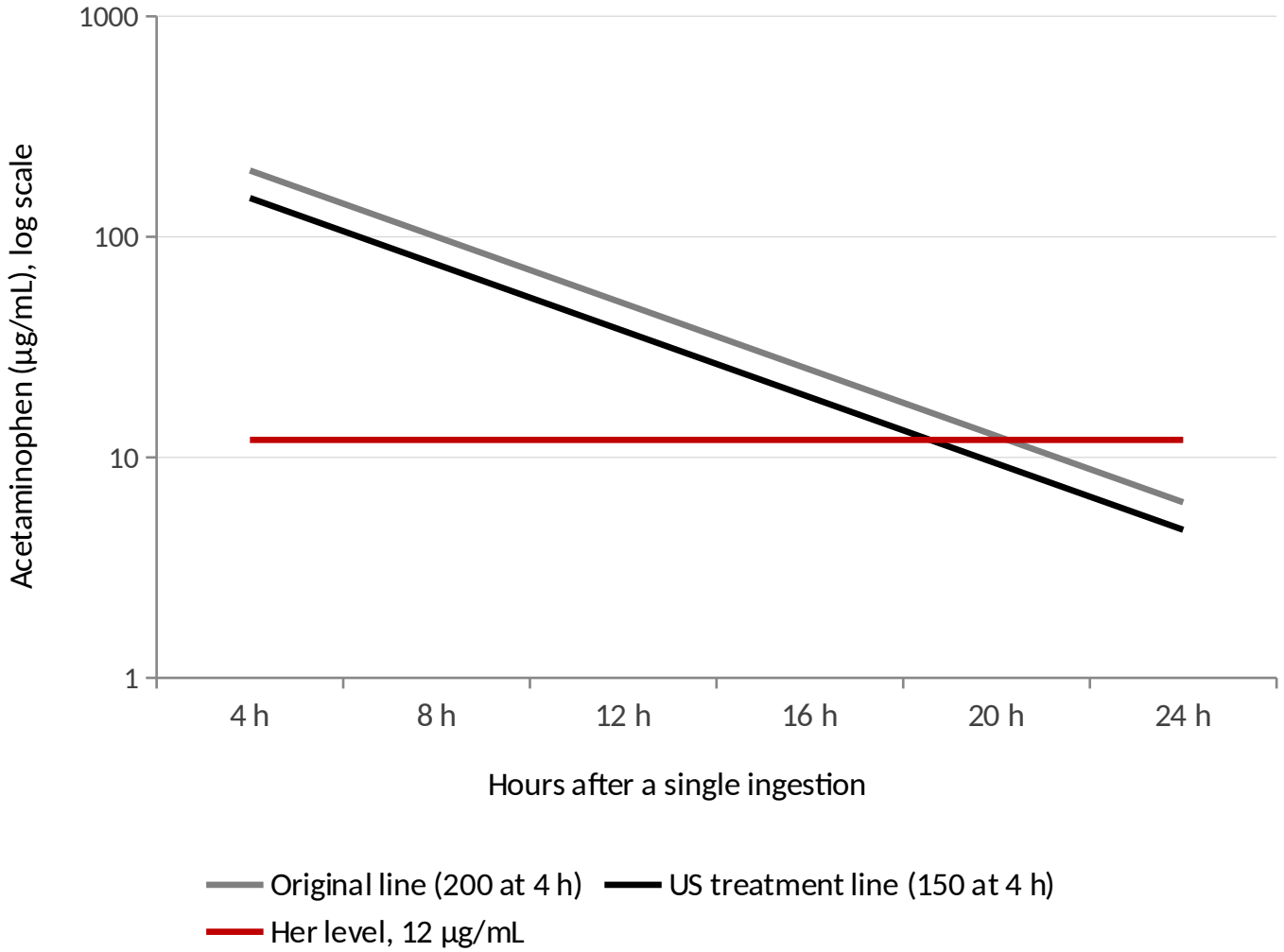
28F, 4 days of nausea

- Roommate: "She was taking a lot of the Tylenol with codeine, plus cold medicine, for 3 days"
- Acetaminophen level: **12 µg/mL**, time of last dose unknown



Can you use the nomogram?

The Rumack-Matthew nomogram



- **One acute ingestion at a known time: it tells you when NOT to give NAC**
- Late, repeated or unknown-time ingestion: a low level doesn't rule out liver injury

If she had taken it all at once

- 8 hours: 12 is below the line, no NAC
- 20 hours: above the line, NAC
- But it was 3 days of repeated doses. **There is no time zero**. Give NAC

Rumack BH, Matthew H. Pediatrics 1975; US/Canada acetaminophen consensus 2023 [PMID: 37552484]

28F, 4 days of nausea

- Roommate: "She was taking a lot of the Tylenol with codeine, plus cold medicine, for 3 days"
- Acetaminophen level: **12 µg/mL**, time of last dose unknown



Can you use the nomogram?

- **No.** Rumack-Matthew is for a single acute ingestion with a known time
- Repeated supratherapeutic use and late presentation: the level can be low or undetectable
- **Start NAC now**

NAC for every kind of acute liver failure?

Published in final edited form as:
Gastroenterology. 2009 September ; 137(3): 856–864.e1. doi:10.1053/j.gastro.2009.06.006.

INTRAVENOUS *N*-ACETYLCYSTEINE IMPROVES TRANSPLANT-FREE SURVIVAL IN EARLY STAGE NON-ACETAMINOPHEN ACUTE LIVER FAILURE

WM Lee[1], **LS Hynan**[1], **L Rossaro**[2], **RJ Fontana**[3], **RT Stravitz**[4], **AM Larson**[5], **TL Davern II**[6], **NG Murray**[7], **T McCashland**[8], **JS Reisch**[1], **PR Robuck**[9], and **the Acute Liver Failure Study Group**

What do you think they found?

NAC for every kind of acute liver failure?

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Gastroenterology. 2009 September ; 137(3): 856–864.e1. doi:10.1053/j.gastro.2009.06.006.

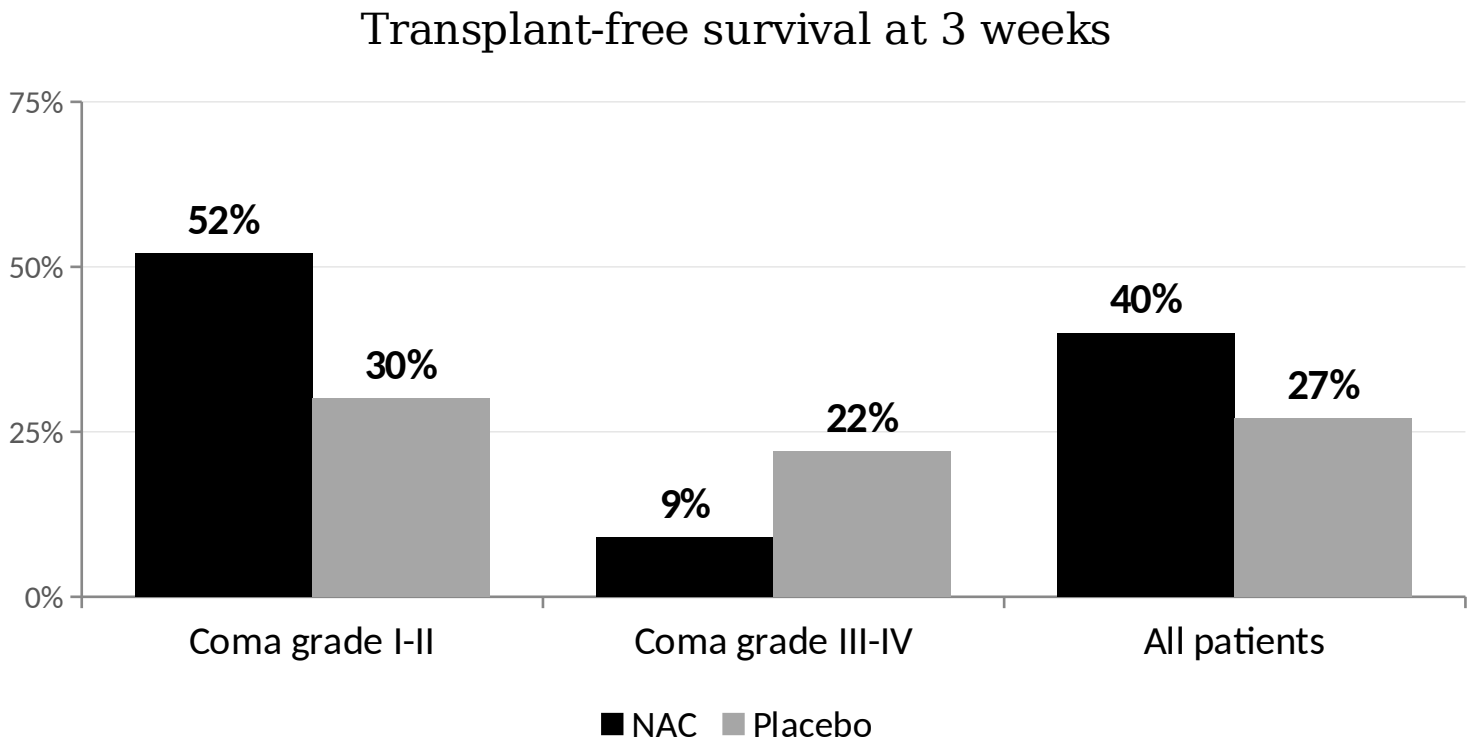
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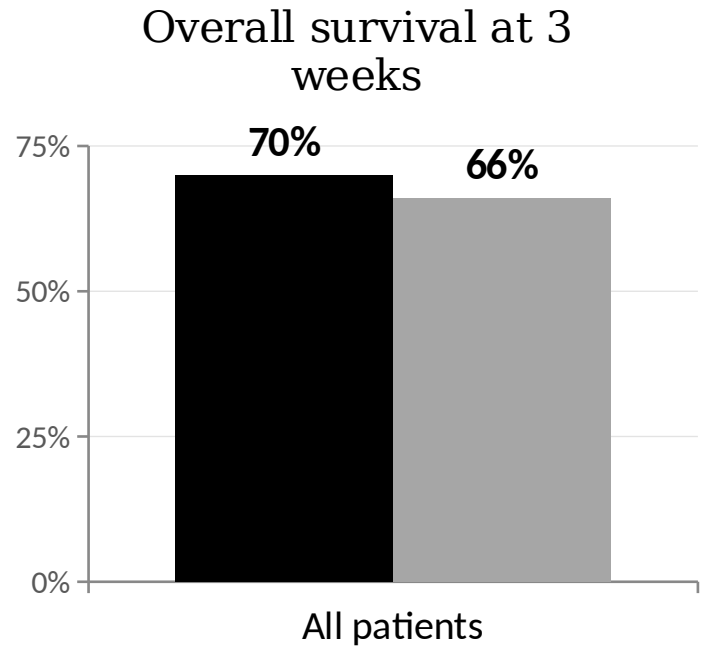
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Conclusions—Intravenous NAC improves transplant-free survival in patients with early stage non-acetaminophen related acute liver failure. Patients with advanced coma grades do not benefit from NAC and typically require emergency liver transplantation. (ClinicalTrials.gov number NCT00004467)

NAC for every kind of acute liver failure?

Adult non-acetaminophen ALF, randomized, NAC vs placebo, 173 patients



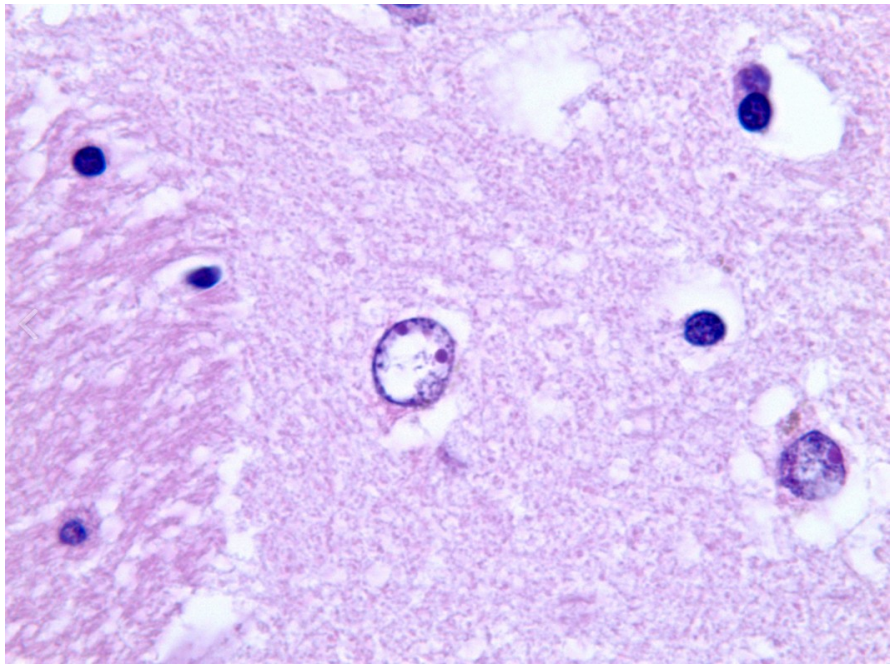


Reasonable in early non-acetaminophen ALF. Don't extrapolate to advanced coma, children or ACLF

Lee WM et al. Gastroenterology 2009 [PMID: 19524577]

Her brain

Hour 12



Alzheimer type II astrocytes: the brain adapting over months of cirrhosis. In ALF there are no months

28F, 4 days of nausea

	Admission	Hour 12
Mental status	Disoriented, asterixis	**Responds only to voice (grade III)**
Ammonia (arterial)	Not sent	**182 µmol/L**
Urine output	OK	15 mL/h for 4 hours
Lactate (after 3 L)	2.1 mmol/L	4.6 mmol/L
Creatinine	1.0 mg/dL	1.3 mg/dL
Sodium	136 mmol/L	133 mmol/L
Glucose	54 mg/dL	Needs D10

What worries you most?

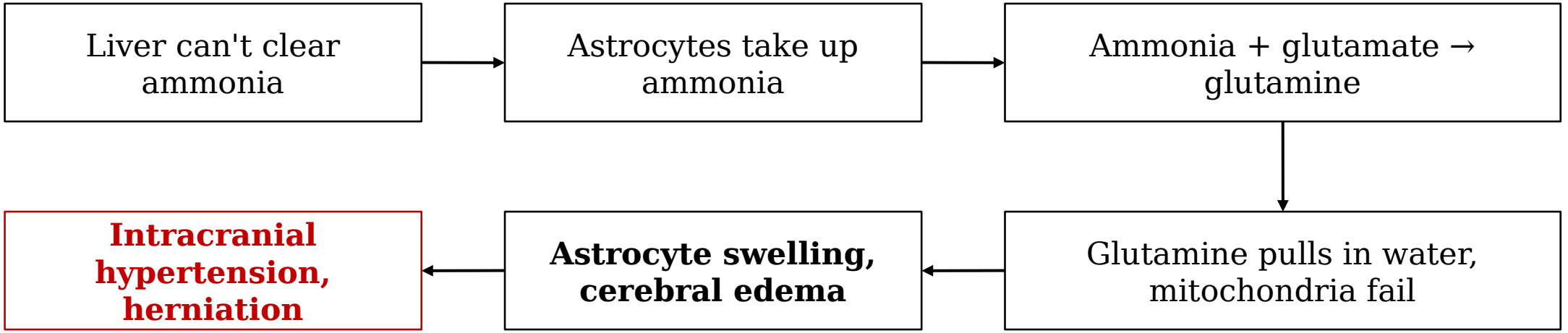
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	Admission	Hour 12
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Sodium	136 mmol/L	133 mmol/L
Glucose	54 mg/dL	Needs D10

Her brain

When do you intubate? Do you wait for the creatinine before dialysis?

Why does the brain swell?



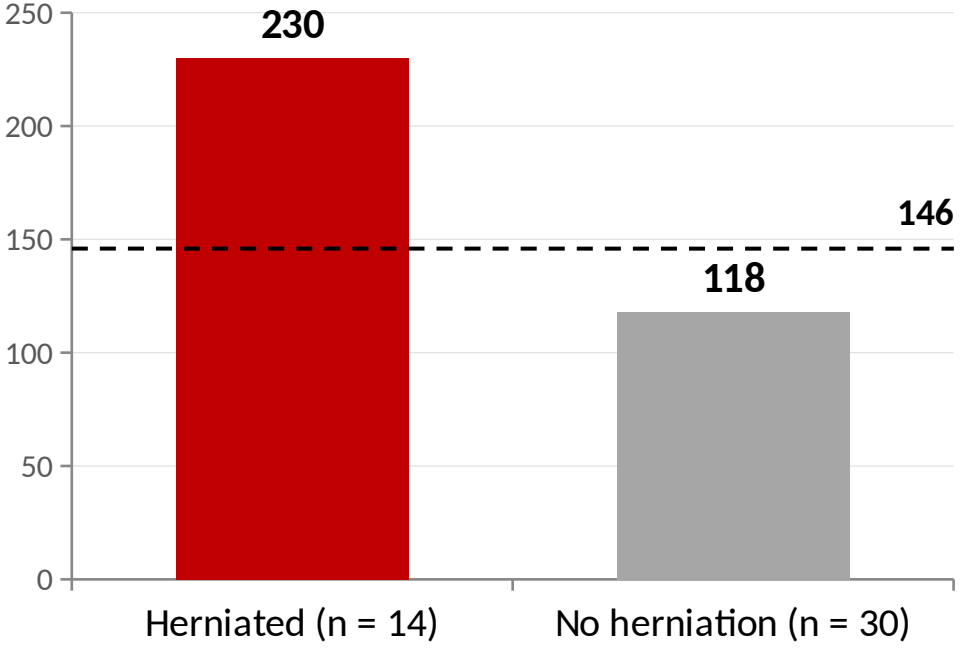
- **Makes it worse:** inflammation and infection, loss of autoregulation, fever, low sodium, high CO2
- **Highest risk:** hyperacute course, young, grade III-IV encephalopathy, ammonia persistently above 150 to 200 µmol/L, AKI, vasopressors

Update 2022 [PMID: 36001996]; practical update 2024 [PMID: 39170946]

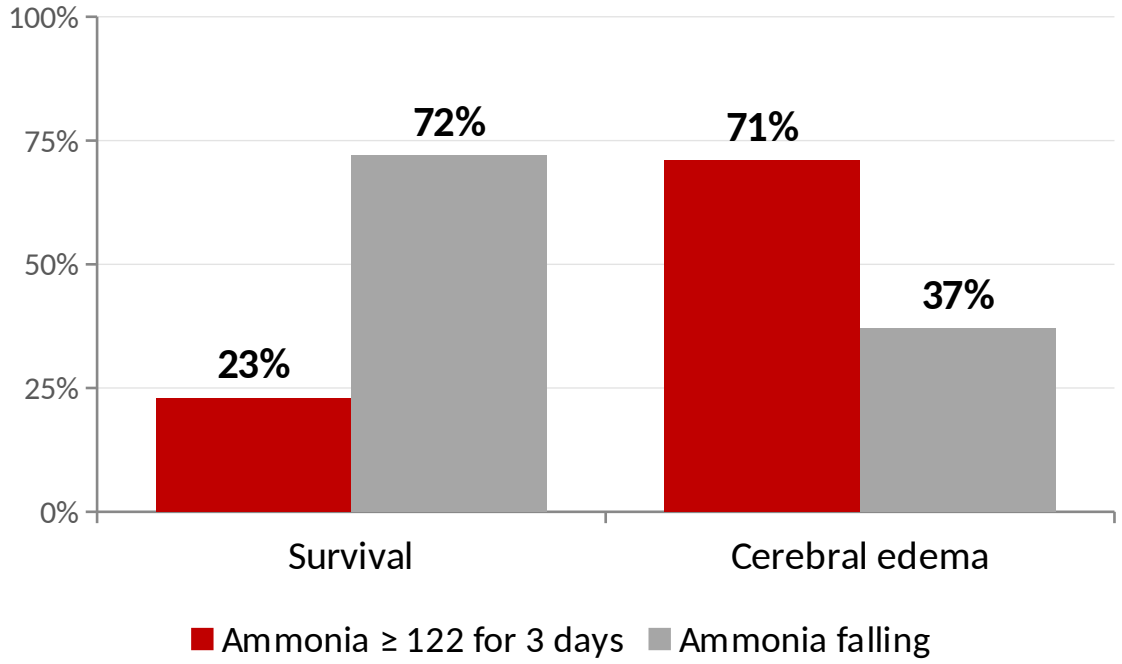


Ammonia: how high is too high?

Clemmesen 1999: ammonia, µmol/L



Kumar 2012: 295 patients



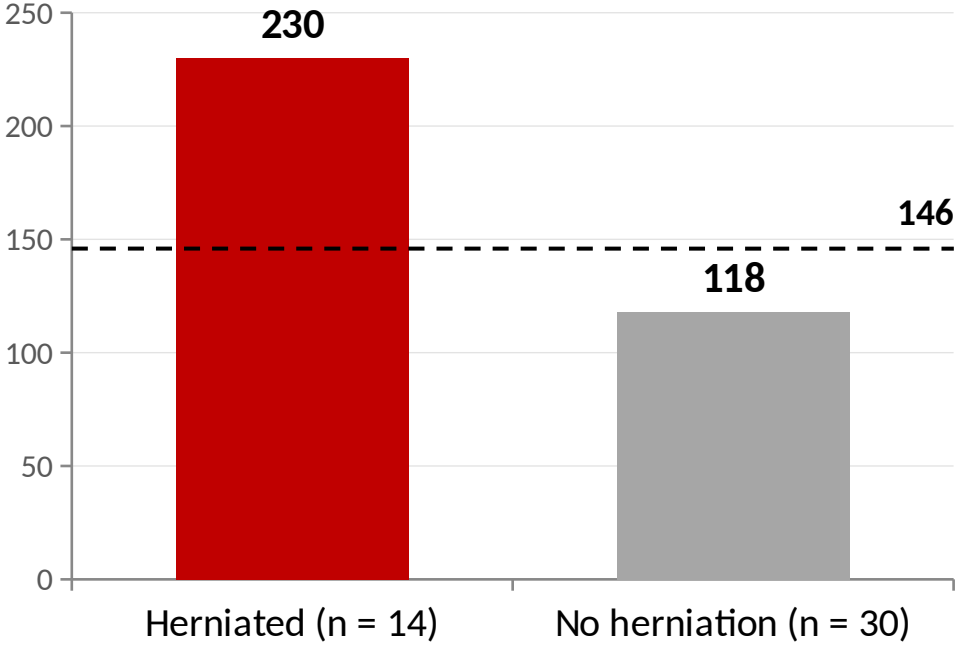
So what number worries you?

Clemmesen JO et al. Hepatology 1999 [PMID: 10051463]; Kumar R et al. Clin Gastroenterol Hepatol 2012 [PMID: 22521861]; more studies in the appendix

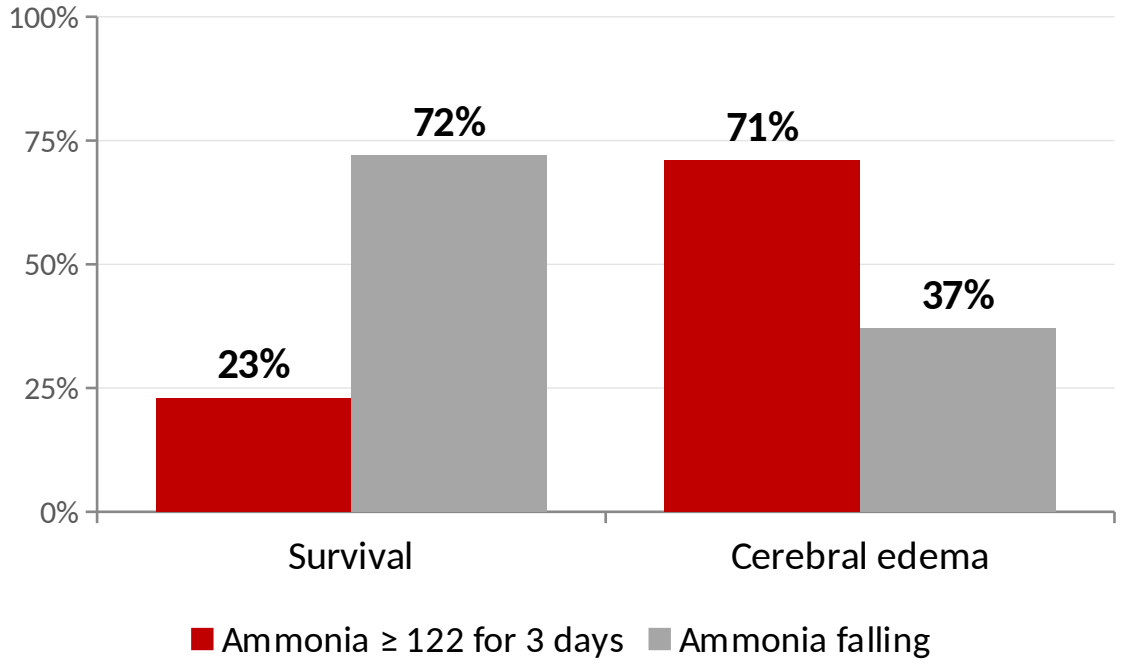


Ammonia: how high is too high?

Clemmesen 1999: ammonia, µmol/L



Kumar 2012: 295 patients



Above 150 to 200 and not falling: worry about the brain

No single level rules ICH in or out. Read it with the HE grade and the trend

Clemmesen JO et al. Hepatology 1999 [PMID: 10051463]; Kumar R et al. Clin Gastroenterol Hepatol 2012 [PMID: 22521861]; more studies in the appendix

Lactulose for her?

Hour 12: ammonia 182 µmol/L, grade III. In a cirrhotic we'd already be giving it

Yes or no?

Lactulose for her?

Hour 12: ammonia 182 µmol/L, grade III. In a cirrhotic we'd already be giving it

Why people give it	Why not
Ammonia drives the brain swelling	**No proven benefit. ALFSG matched comparison: a little longer survival time, no difference in encephalopathy or outcome**
It's what we do in cirrhosis	Gas-distended bowel gets in the surgeon's way at transplant
Cheap, and the AASLD position paper allows it in early grades, short of diarrhea	Diarrhea means free water loss and volume swings, which fight your sodium target
	Liquid by mouth or tube in a patient who can't protect her airway: aspiration

Not for her: CRRT for the ammonia, protect the airway, keep the sodium up

AASLD position paper 2011, rec. 26; Alba L, Hay JE, Lee WM et al. J Hepatol 2002;36(Suppl 1):33 (abstract); ACG ALF 2023 [PMID: 37377263]

When do you intubate?

- Responds to voice, drifts off, intermittently agitated
- Handling secretions for now
- SpO2 96% on 2 L

Intubate now, or wait until she can't protect her airway?

When do you intubate?

Grade III-IV encephalopathy in ALF: intubate

Before	During	After
Preoxygenate Norepinephrine ready Check glucose Head of bed 30°, neck neutral	Avoid hypotension (perfusion pressure) Avoid coughing and bucking (ICP spikes) Most experienced operator	Propofol: short-acting, may lower ICP Avoid long-acting benzodiazepines Aim for normal CO2, check a gas Sedation hides the neuro exam: check pupils

Practical update 2024 [PMID: 39170946]; ACG ALF 2023 [PMID: 37377263]; SCCM 2020 [PMID: 32058387]

Can we use permissive hypercapnia?

After intubation: bilateral infiltrates, P/F 150 on PEEP 10

Lung protection or brain protection?

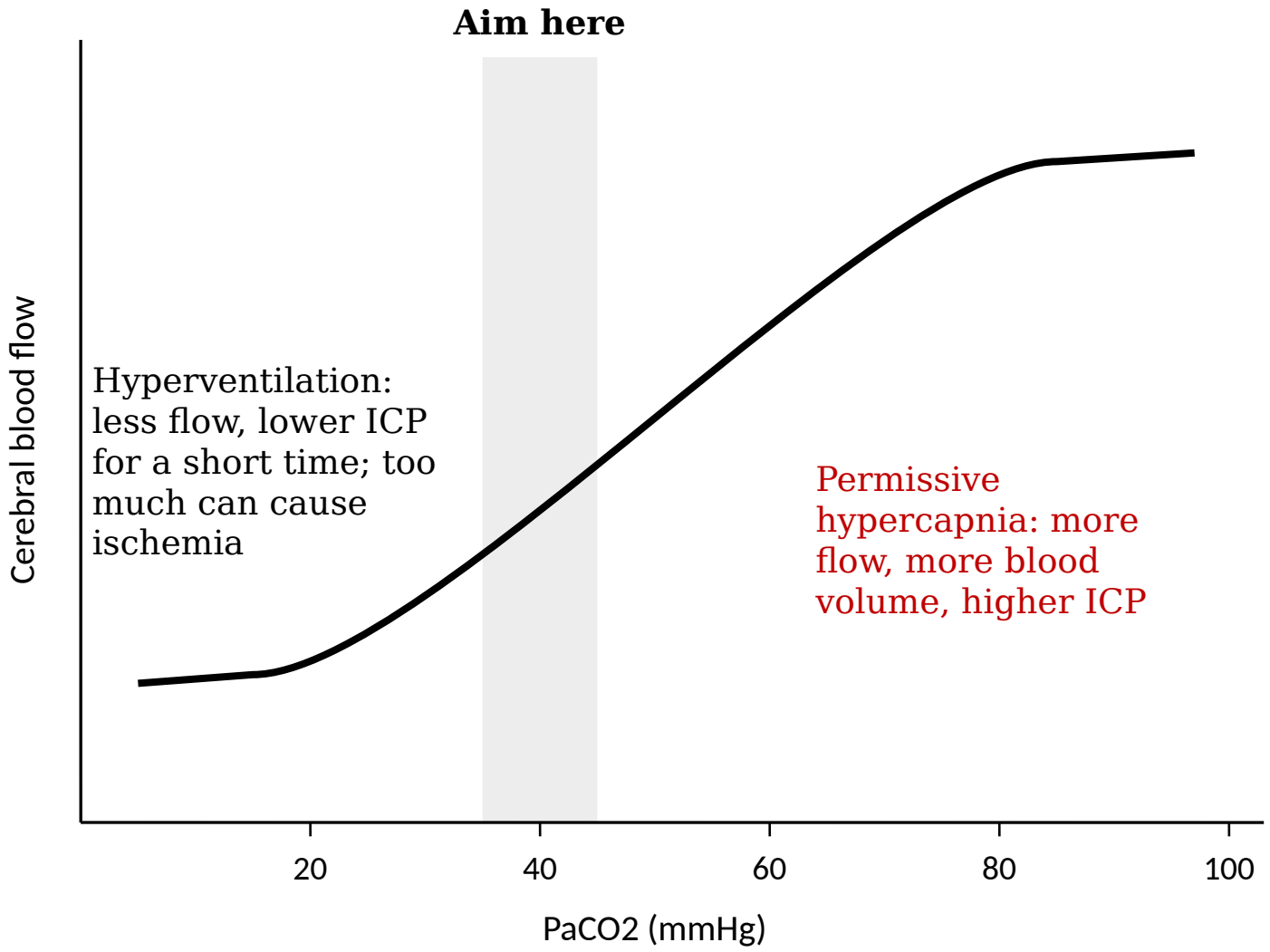
Can we use permissive hypercapnia?

After intubation: bilateral infiltrates, P/F 150 on PEEP 10

	The lung wants	The brain wants	What we do
Tidal volume	6 mL/kg predicted body weight	Fine, but low volumes can push the CO2 up	Do it, and raise the rate to hold the CO2
CO2	Tolerate a high CO2	High CO2 dilates vessels and raises ICP	**Aim for normal CO2. No permissive hypercapnia when ICP is a concern**
Hyperventilation	No	Lowers ICP briefly; if overdone or prolonged, can cause ischemia	Only as a short rescue for herniation
PEEP	Recruit	Very high pressures can impair venous return	Set it for oxygen and blood pressure. PEEP is not forbidden

SCCM 2020 [PMID: 32058387]; practical update 2024 [PMID: 39170946]

CO2 and cerebral blood flow

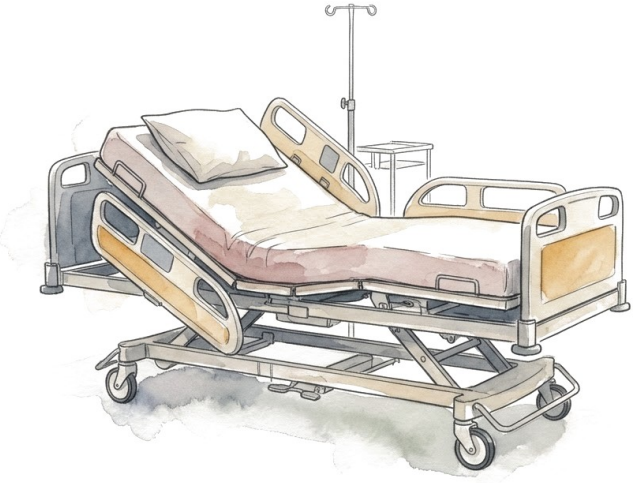


- Between about 20 and 80 mmHg, cerebral blood flow rises with PaCO2
- In a swollen brain, more blood volume means a higher ICP
- Keep PaCO2 normal. Hyperventilate only as a short bridge for herniation

Schematic, not to scale. Standard cerebral physiology

Protecting the brain: every patient with grade III-IV

- Head of bed 30°, neck neutral
- Treat fever, avoid hypoglycemia
- Sodium **145 to 155 mmol/L** with hypertonic saline (use your center's protocol)
- Avoid fluid overload
- Treat seizures, EEG if in doubt. No routine prophylaxis
- No cooling: 34 °C did not prevent intracranial hypertension in an RCT



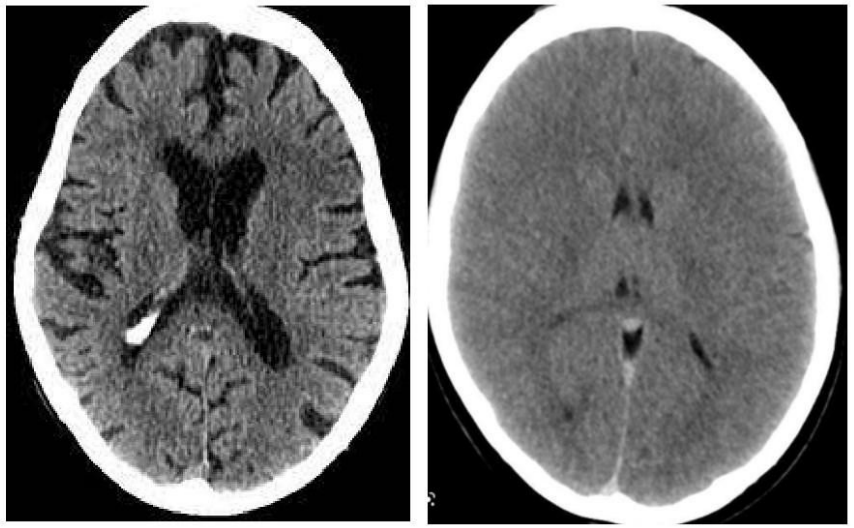
SCCM 2023 [PMID: 37052436]; update 2022 [PMID: 36001996]; Bernal W et al. J Hepatol 2016;65:273-9

A surge: new hypertension, bradycardia or a pupil

- Hyperosmolar bolus (hypertonic saline or mannitol)
- Brief hyperventilation as a bridge
- Call the transplant team now

Things to know

- Hypertension and bradycardia come late. Don't wait for them
- A normal head CT does not exclude raised ICP
- ICP monitors are not routine; SCCM suggests against (very low certainty)



Normal | Effacement of sulci

SCCM 2023 [PMID: 37052436]; practical update 2024 [PMID: 39170946]

The creatinine is only 1.3

Ammonia 182 → 214 µmol/L, grade III, urine output 15 mL/h

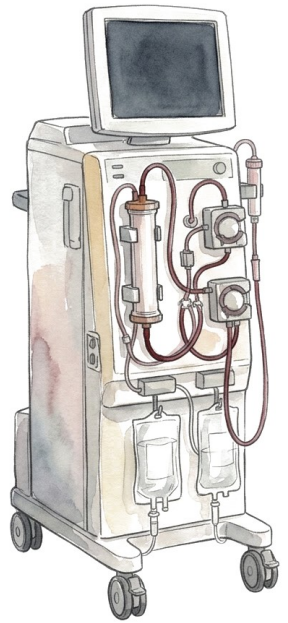
Do you wait for a conventional dialysis indication?

The creatinine is only 1.3

Ammonia 182 → 214 µmol/L, grade III, urine output 15 mL/h

No. In ALF, start CRRT for the ammonia

- ACG 2023: suggests early CRRT for hyperammonemia with grade II or higher encephalopathy, even without a renal indication (conditional, very low certainty)
- Continuous, not intermittent: fewer swings in blood pressure and brain water
- Call nephrology now



ACG ALF guideline 2023 [PMID: 37377263]; update 2022 [PMID: 36001996]

Should we correct this INR?

Hour 12: INR 4.1, platelets 88,000. She needs a dialysis catheter

2 units of FFP first?

Should we correct this INR?

Hour 12: INR 4.1, platelets 88,000. She needs a dialysis catheter

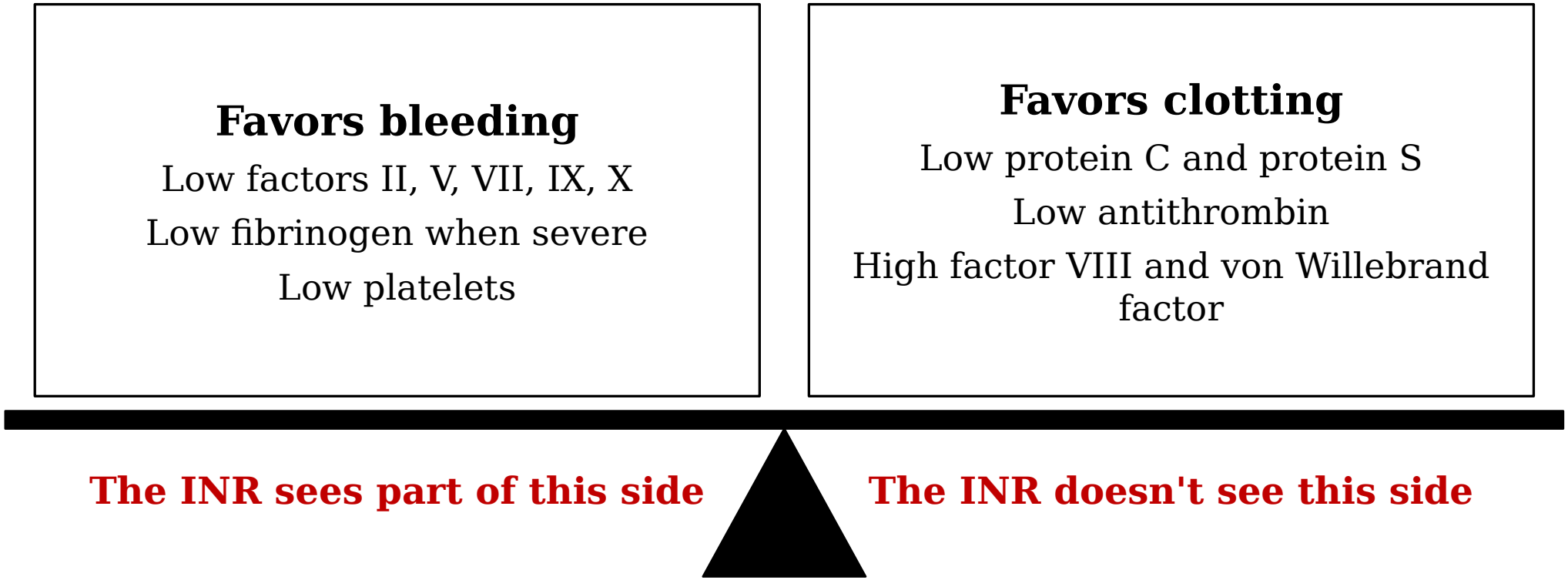
- **No plasma just to fix the number**
- The INR sees only the procoagulant side. Protein C and S are low too, factor VIII and von Willebrand factor are high. Most patients are rebalanced
- Plasma adds volume (lungs, brain) and erases the INR trend that prognosis and listing rely on

What to do instead

- Ultrasound-guided line, experienced operator, compressible site
- Vitamin K once; check fibrinogen and platelets; viscoelastic testing if available
- Treat bleeding. Plan high-risk procedures (like an ICP monitor) separately

Update 2022 [PMID: 36001996]; SCCM 2020 [PMID: 32058387]

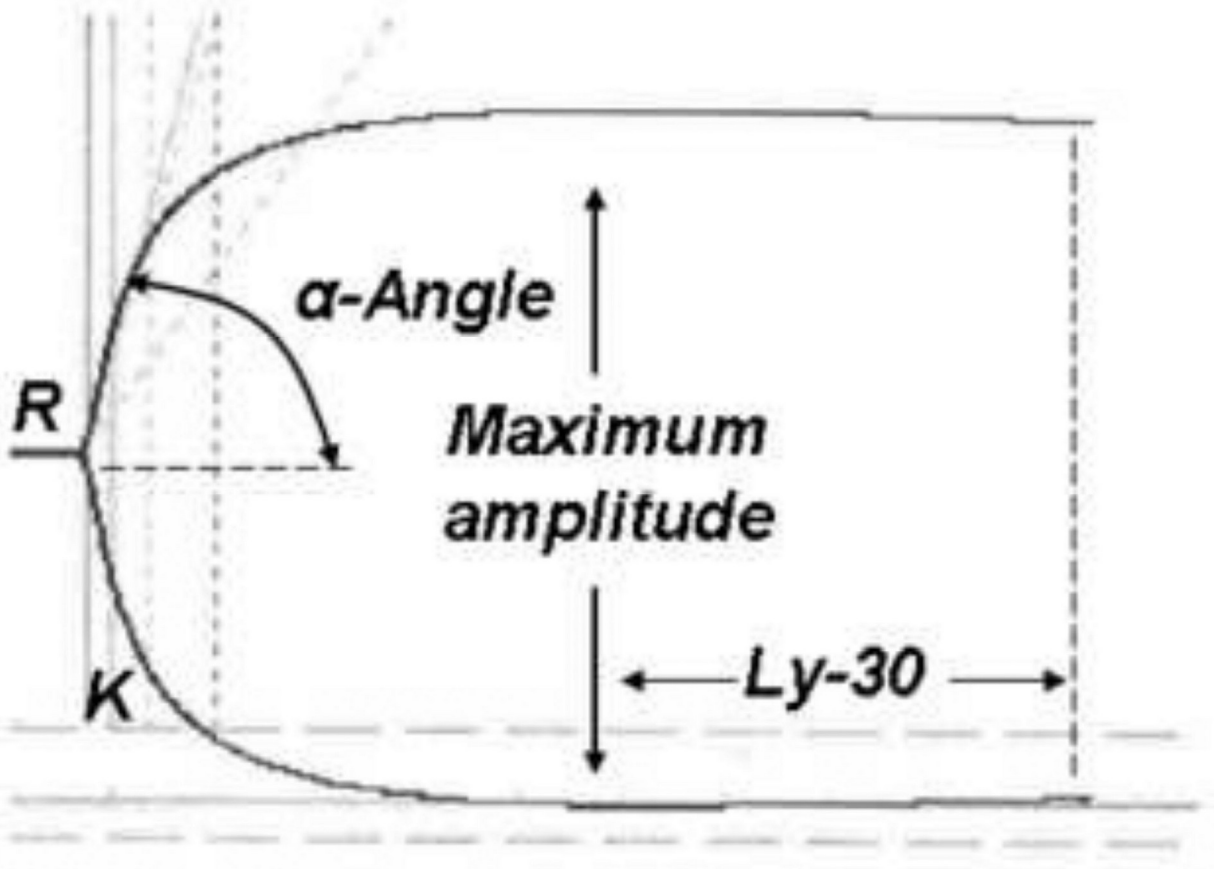
Rebalanced hemostasis



Balanced in most patients, but easily tipped either way: both bleeding and thrombosis happen

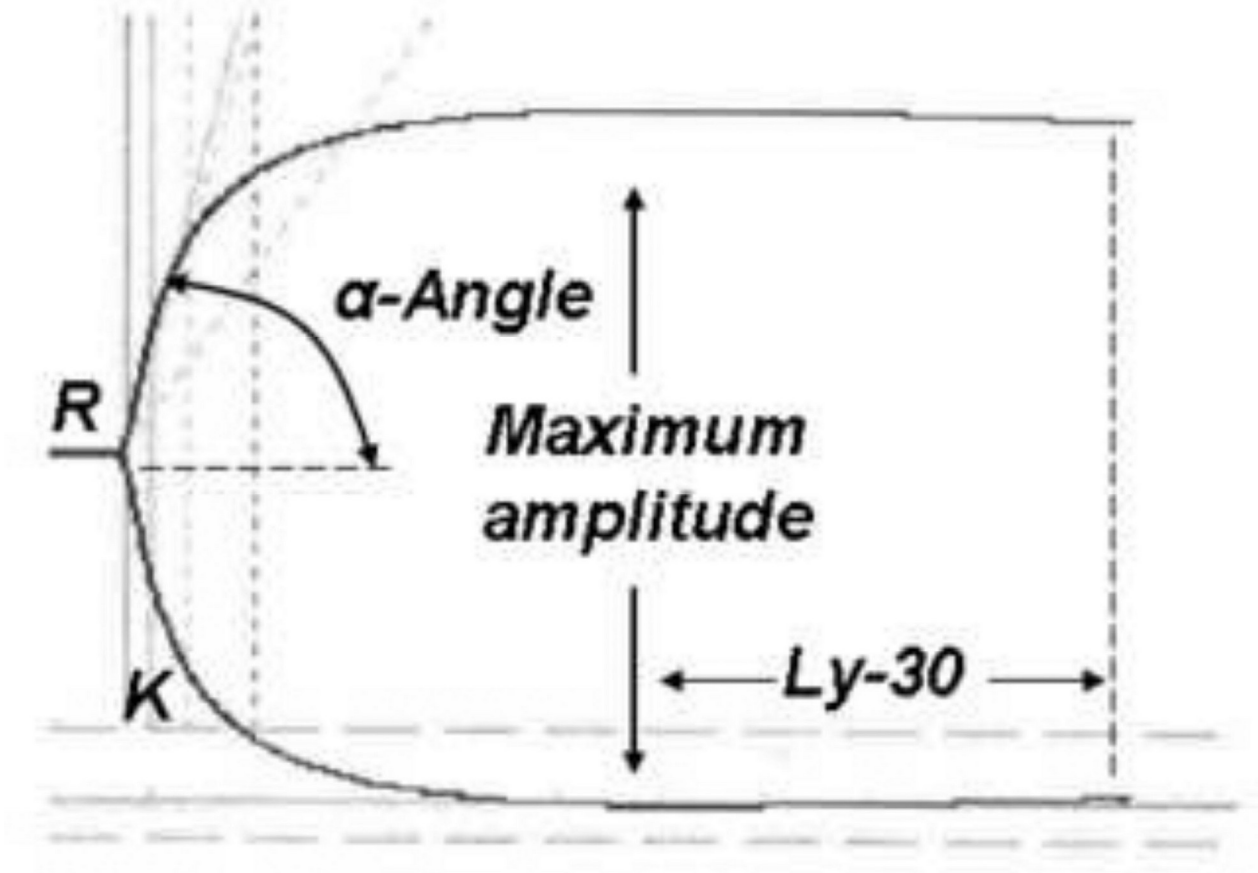
Concept from Lisman T, Porte RJ. Blood 2010

INR 4.2. What does the TEG say?



Stravitz RT et al. J Hepatol 2012;56:129-36 [PMID: 21703173]

INR 4.2. What does the TEG say?



Normal in all five TEG measures, in a patient with ALF and an INR of 4.2

Stravitz RT et al. J Hepatol 2012;56:129-36 [PMID: 21703173]

CRRT for ammonia: what to agree on with nephrology

	What to watch and say
Why	It's for the ammonia and the brain, not the creatinine. Give them the HE grade, the ammonia trend, urine output and fluid status
Dose	Prescribed vs delivered. Filter clotting and downtime mean less clearance, and the ammonia climbs back. Recheck it
Sodium	One plan for replacement fluid or dialysate, hypertonic saline and every other sodium source. Check the sodium often
Safety	Phosphate and potassium fall. Blood pressure drops at connection. Anticoagulation per your protocol; citrate can build up in liver failure, so follow the total to ionized calcium ratio

ACG ALF 2023 [PMID: 37377263]; Cardoso FS et al. Hepatology 2018 [PMID: 28859230]; update 2022 [PMID: 36001996]

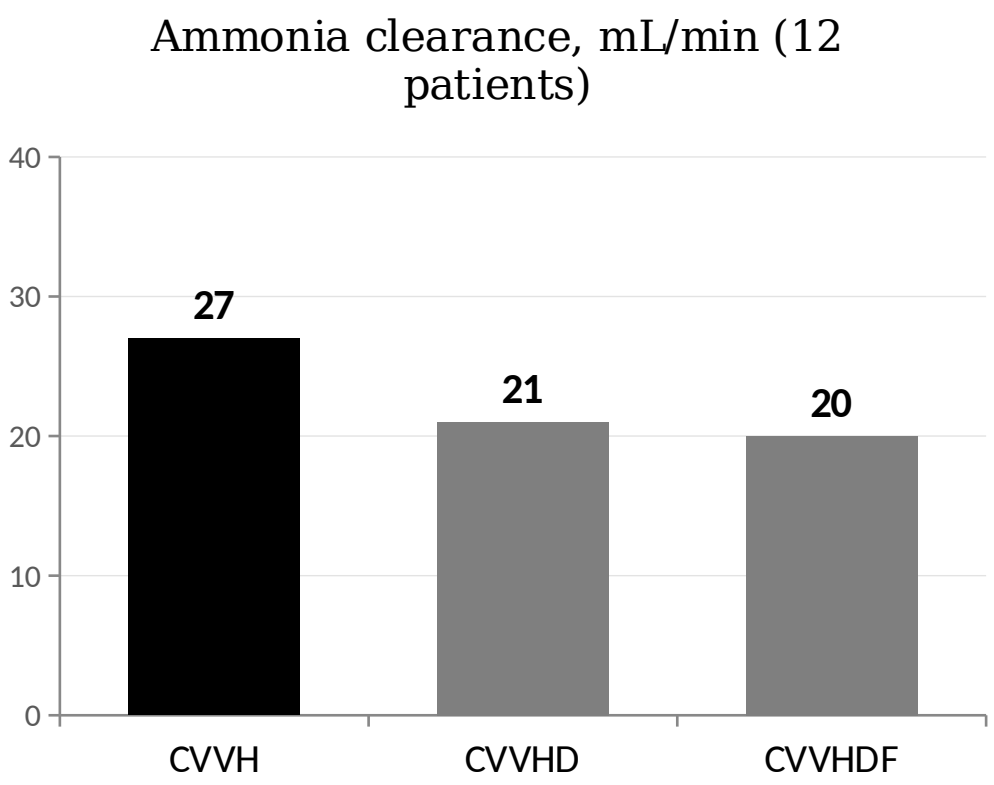
I know what you're thinking...

"CVVH doesn't clear ammonia"

True or false?

"CVVH doesn't clear ammonia"

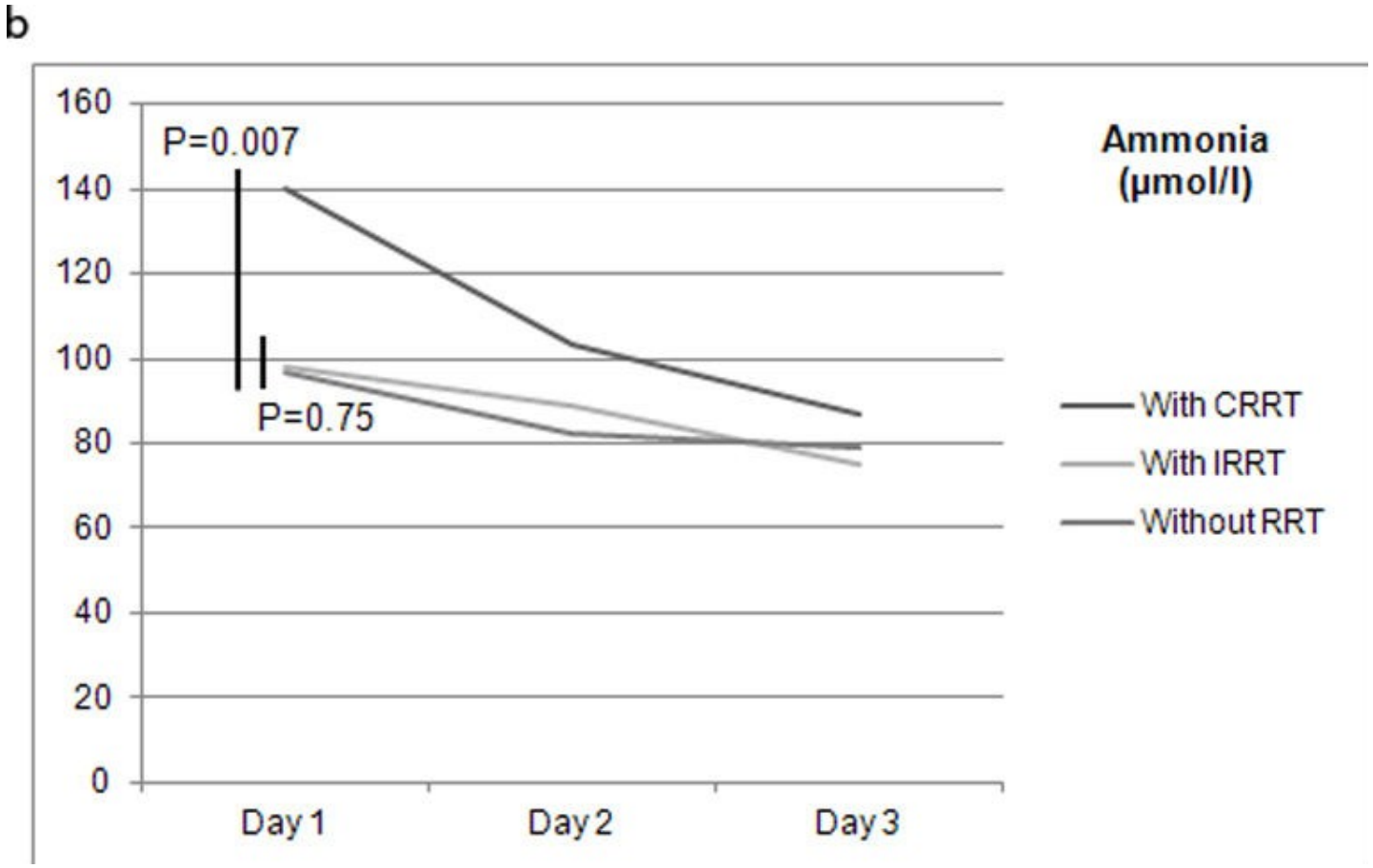
- Ammonia is tiny (17 Da), water soluble and not protein bound
- It crosses the filter with the water (convection, CVVH) or down its gradient (diffusion, CVVHD)
- So what matters is effluent flow and hours on a working circuit, not the letters
- More effluent, more clearance: **90 vs 35 mL/kg/h** (Slack 2014)
- 54 ALF patients on CRRT: median ammonia **151 → 52 µmol/L** by day 5 (Warrillow 2020)



It does. About half the clearance of urea, so dose and filter life matter

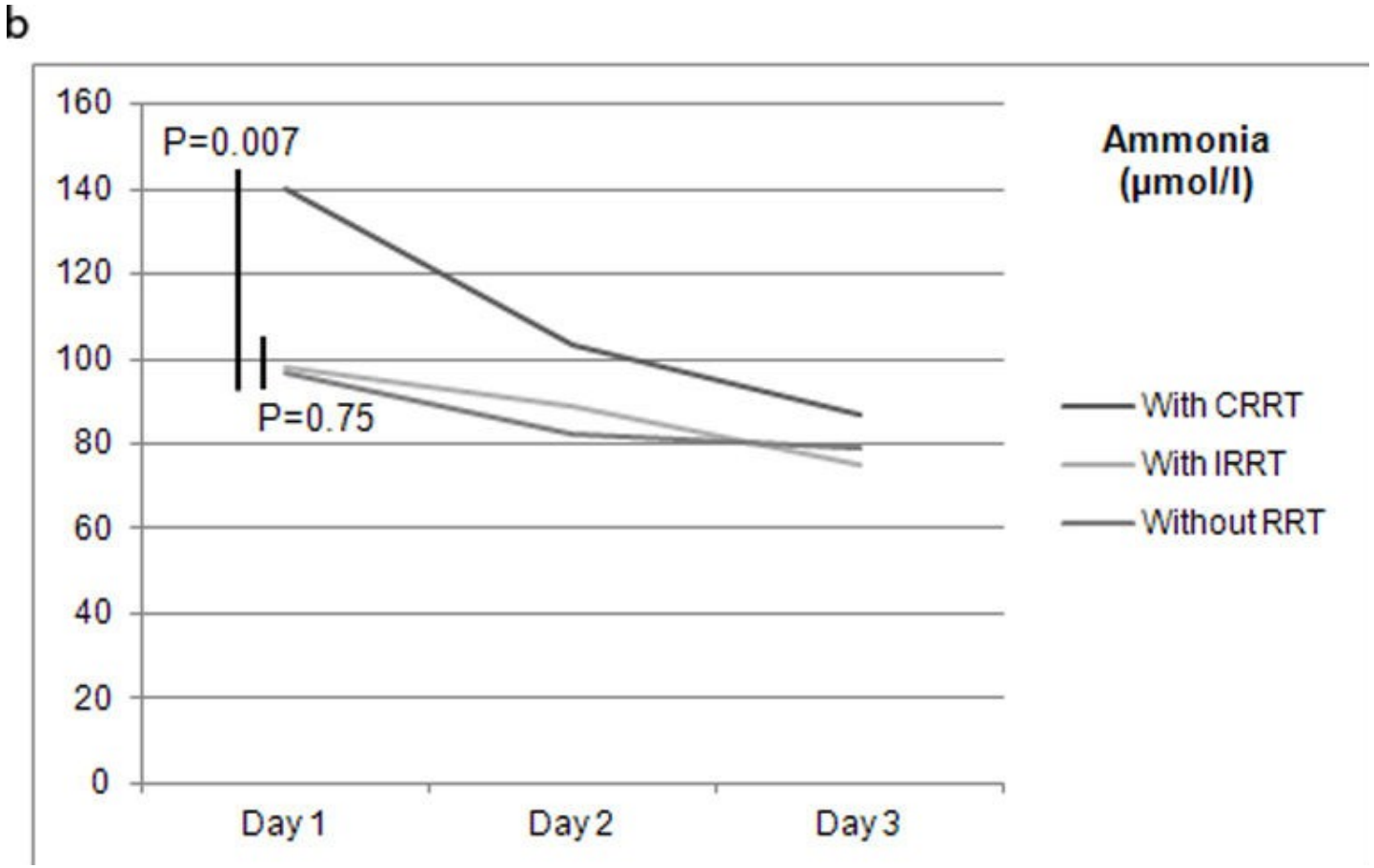
Fisher C et al. Blood Purif 2022 [PMID: 35042216]; Warrillow SJ et al. Crit Care Med 2020 [PMID: 31939790]; Slack AJ et al. Liver Int 2014

Does early CRRT actually help?



Cardoso FS et al. Hepatology 2018;67:711-20 [PMID: 28859230]

Does early CRRT actually help?



Ammonia fell from about 140 to under 90 µmol/L over 3 days on CRRT. Little change with intermittent RRT or no RRT

Cardoso FS et al. Hepatology 2018;67:711-20 [PMID: 28859230]

Does early CRRT actually help?

	US ALF Study Group cohort (2018)	**Pilot RCT (2026)**
Design	Large registry, observational	Open label, ALF with cerebral edema; both arms got plasma exchange
Comparison	CRRT vs no CRRT	CRRT within 12 hours vs CRRT only if plasma exchange failed
Finding		
Catch		

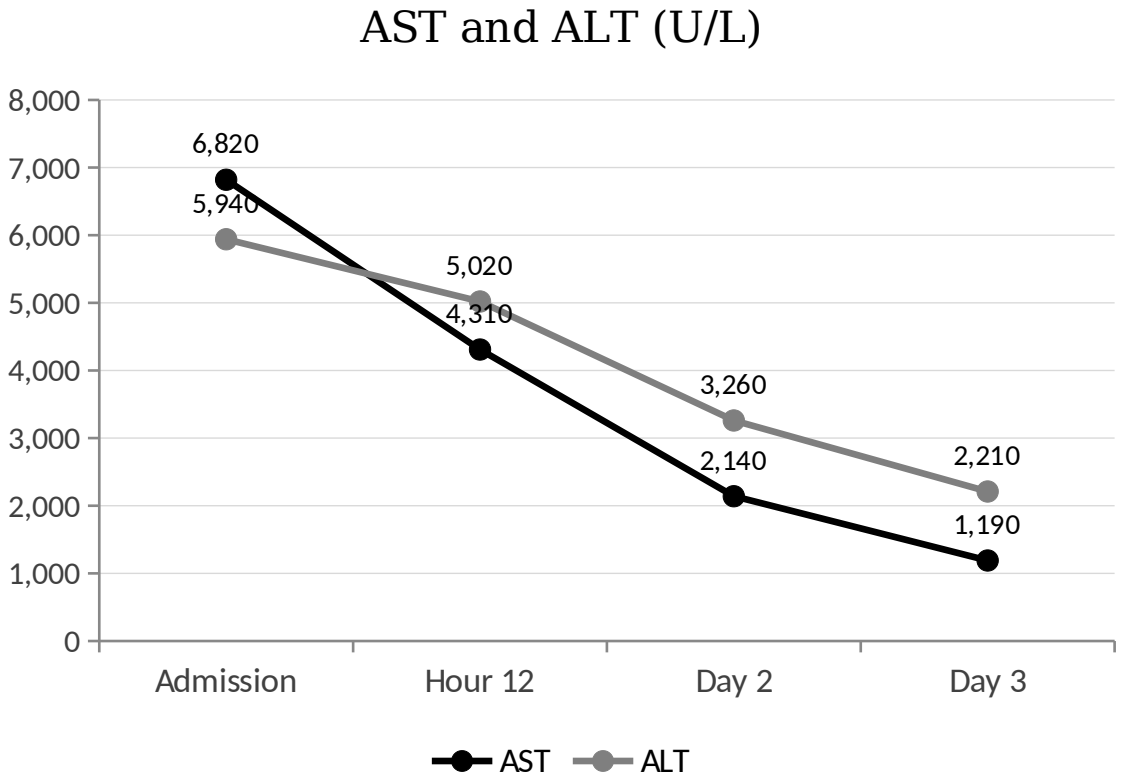
Cardoso FS et al. Hepatology 2018 [PMID: 28859230]; Aliment Pharmacol Ther 2026 [PMID: 42332991]

Does early CRRT actually help?

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Design	Large registry, observational	Open label, ALF with cerebral edema; both arms got plasma exchange
Comparison	CRRT vs no CRRT	CRRT within 12 hours vs CRRT only if plasma exchange failed
Finding	CRRT linked to lower ammonia and better 21-day transplant-free survival	Primary endpoint, 28-day survival: similar by intention to treat. Fewer deaths in the first 7 days (2.2% vs 17.8%)
Catch	**Association. Sicker or healthier patients may have been picked for CRRT**	**Small pilot, one center, many protocol violations. Needs a larger trial**

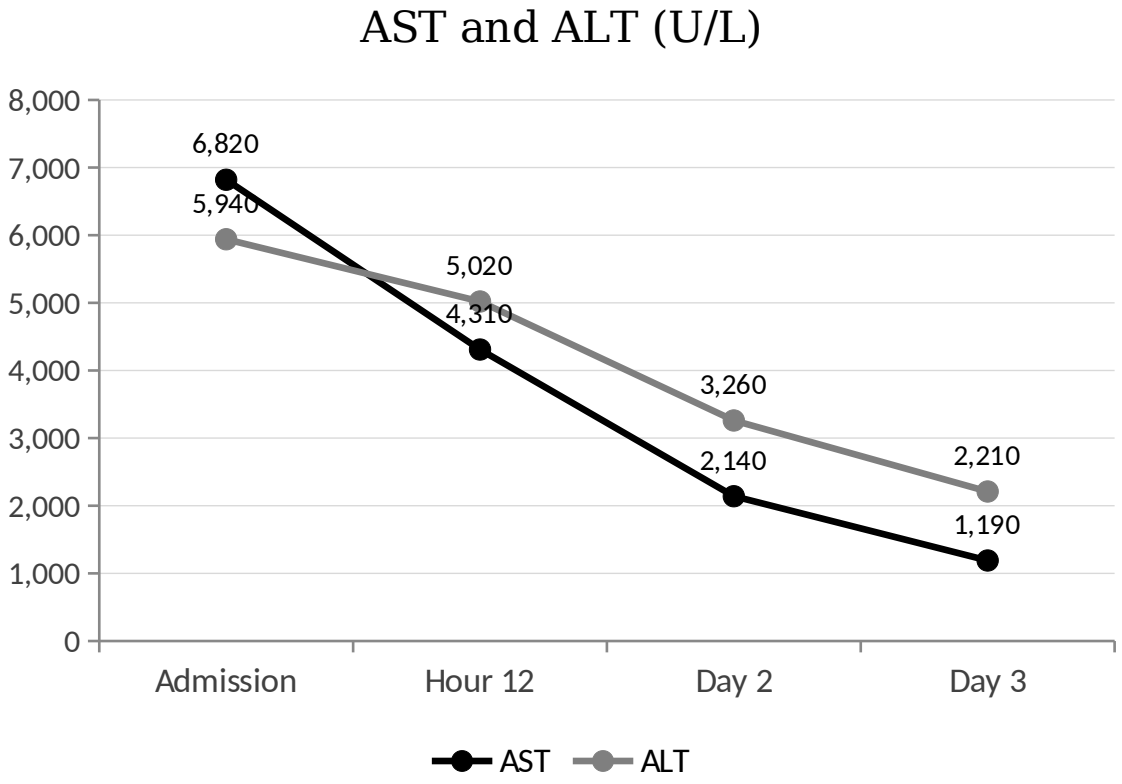
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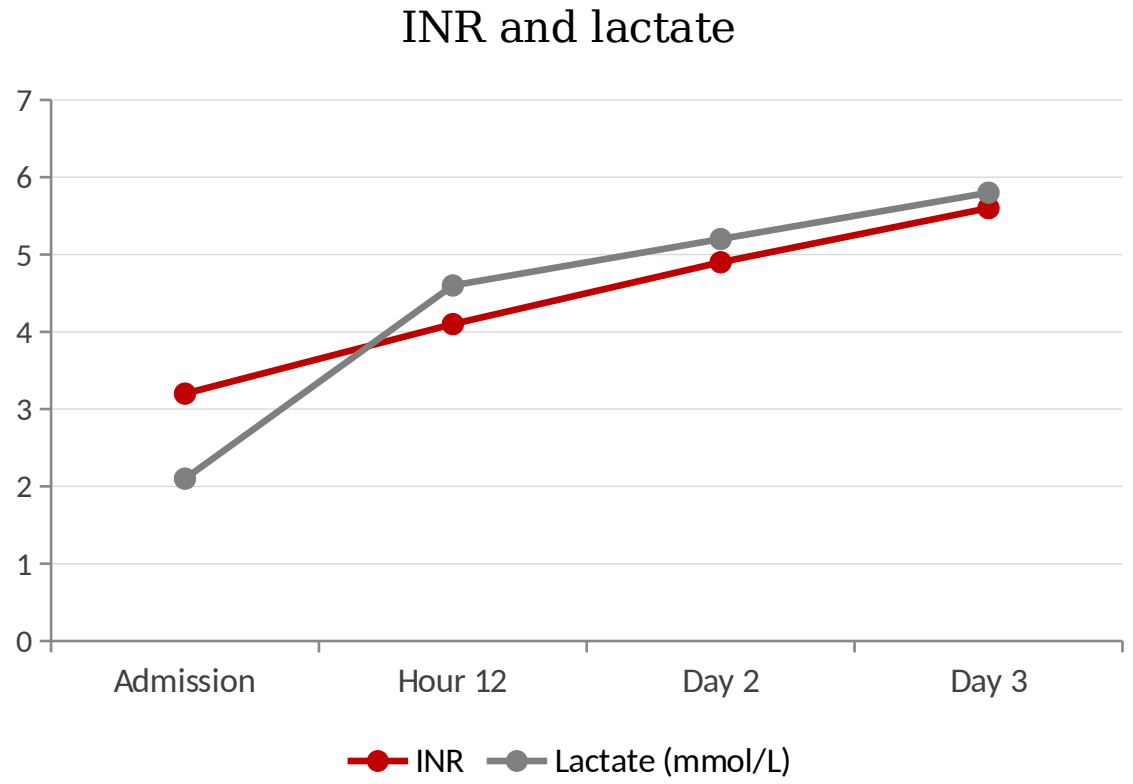
Day 3: the enzymes are falling



Is she getting better?

Day 3: the enzymes are falling





Day 2 arterial pH 7.26 after fluids

No. The INR, the lactate and her brain are all worse

Made-up numbers for teaching

When do you call transplant hepatology?

When she meets King's College criteria?

When MELD is high enough?



When do you call transplant hepatology?

You should have called on admission, when you first suspected ALF

- King's College criteria predict death without transplant. They are specific, not sensitive
- They are not a referral threshold. If you wait for them, the evaluation starts too late
- Patient A: arterial pH **7.26** on day 2 after fluids. She now meets the acetaminophen criteria



O'Grady 1989 criteria, as summarized in EASL 2017 [PMID: 28417882]; full criteria in the appendix

Can plasma exchange buy time?

Research Article

JOURNAL OF HEPATOLOGY

High-volume plasma exchange in patients with acute liver failure: An open randomised controlled trial

Fin Stolze Larsen[1,*], Lars Ebbe Schmidt[1], Christine Bernsmeier[2], Allan Rasmussen[3], Helena Isoniemi[4], Vishal C. Patel[2], Evangelos Triantafyllou[2], William Bernal[2], Georg Auzinger[2], Debbie Shawcross[2], Martin Eefsen[1], Peter Nissen Bjerring[1], Jens Otto Clemmesen[1], Krister Hockerstedt[4], Hans-Jørgen Frederiksen[5], Bent Adel Hansen[1], Charalambos G. Antoniades[2,6,†], Julia Wendon[2,†]

What do you think they found?

Larsen FS et al. J Hepatol 2016;64:69-78 [PMID: 26325537]

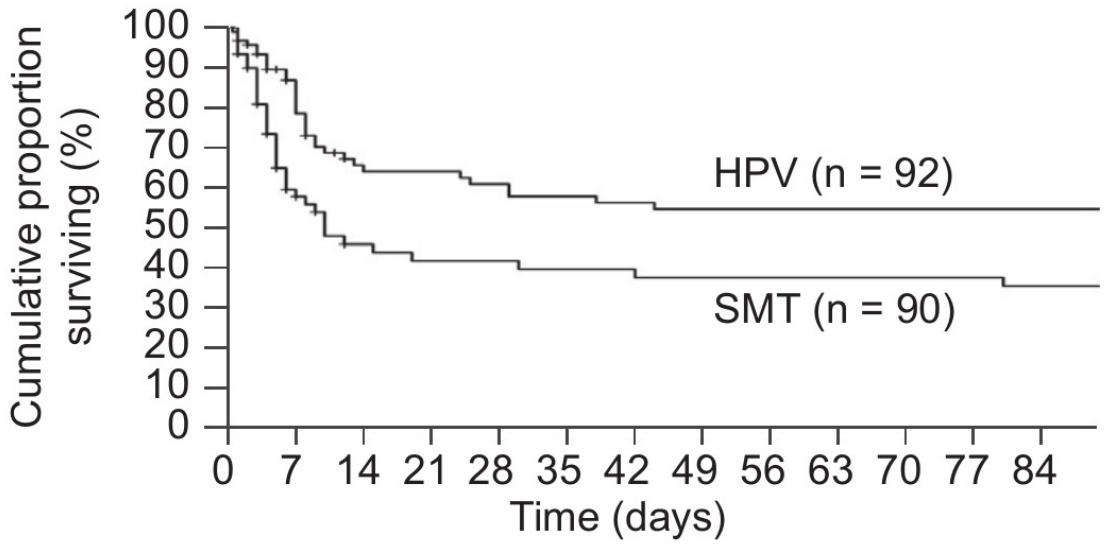
Can plasma exchange buy time?

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Can plasma exchange buy time?

Study	Open randomized trial, ALF, high-volume plasma exchange for 3 days vs standard care, 182 patients
Main finding	Hospital survival 58.7% vs 47.8% (HR 0.56); best in patients who were not transplanted
What it doesn't show	
So	

Larsen FS et al. J Hepatol 2016 [PMID: 26325537]

Can plasma exchange buy time?

Study	Open randomized trial, ALF, high-volume plasma exchange for 3 days vs standard care, 182 patients
Main finding	Hospital survival 58.7% vs 47.8% (HR 0.56); best in patients who were not transplanted
What it doesn't show	**Not blinded; one regimen in one population. Says nothing about ACLF or about albumin dialysis devices**
So	A bridge to discuss with transplant hepatology. Not a reason to delay listing

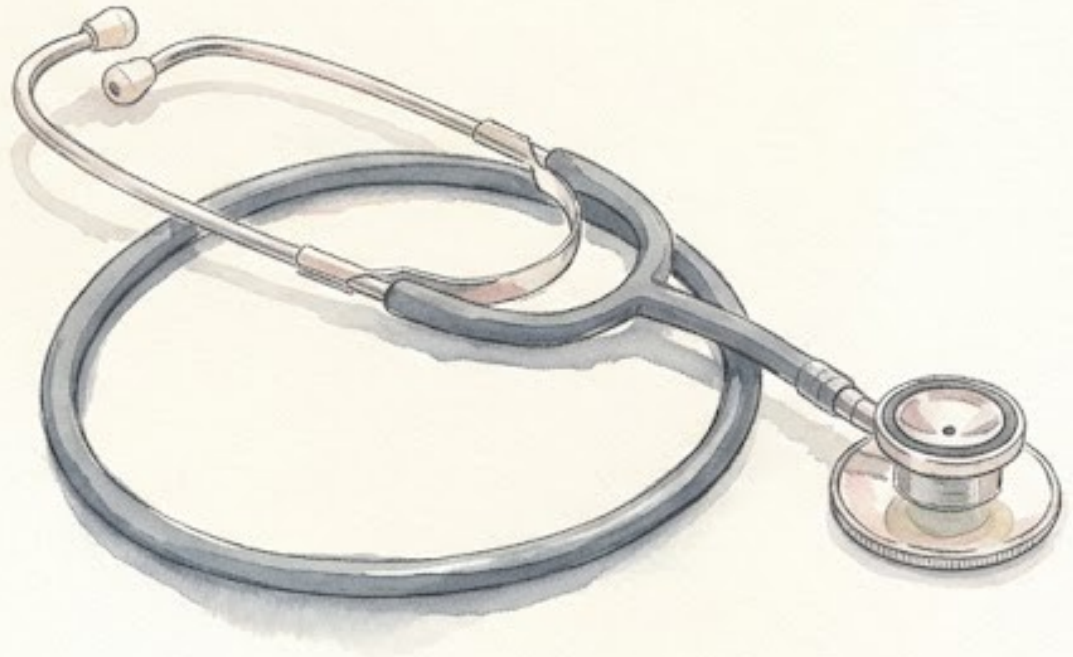
Larsen FS et al. J Hepatol 2016 [PMID: 26325537]

Let's recap: bed 4, day 3

Problem	What she's on	What we watch
Acetaminophen ALF, meets King's (pH 7.26)	NAC. Listed status 1A on day 2	Time to an offer
Grade III-IV encephalopathy, ammonia 182	Intubated, normal CO2, sodium 145 to 155, CRRT	Pupils, ammonia on CRRT, sodium
Lactate 5.8 and rising	Volume and MAP, look for infection	Lactate, MAP, pressor need
INR 5.6, not bleeding	No plasma	Bleeding, INR trend
Infection risk	Surveillance cultures	New fever or shock: low bar for antibiotics and antifungals

Made-up numbers for teaching

Any questions so far?



Bed 7

56M, cirrhosis and fever



Back to day 1: bed 7

Bed 4	Bed 7
28-year-old woman	56-year-old man
Acetaminophen ALF	Alcohol-related cirrhosis with ascites
Intubated, on CRRT, transplant hepatology involved	Fever, confusion, low blood pressure

Is this liver failure too?

56M, cirrhosis and fever

- Alcohol-related cirrhosis, ascites on diuretics; last drink 3 weeks ago
- Temp 38.4 °C, HR 112, BP 82/46
- RR 26, SpO2 92% on a 35% Venturi mask; PaO2 74
- Confused, asterixis (grade II)
- Urine output 15 mL/h

	Now	Baseline
Bilirubin (mg/dL)	5.8	2.1
INR	2.1	1.6
Creatinine (mg/dL)	2.6	0.9
Sodium (mmol/L)	127	133
WBC (/µL)	16,200	
Platelets (/µL)	62,000	
Lactate (mmol/L)	3.9	

What's your next step?

56M, cirrhosis and fever

- Alcohol-related cirrhosis, ascites on diuretics; last drink 3 weeks ago
- Temp 38.4 °C, HR 112, BP 82/46
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What's your next step?

Find the source: cultures, antibiotics, a diagnostic tap

What tipped him over?

INR 2.1, platelets 62,000. Do you tap him?

What tipped him over?

- **Yes, now. No plasma, no platelets**
- Ascitic fluid: **PMN 1,240/mm³**. Spontaneous bacterial peritonitis



Common triggers

- Infection: SBP, pneumonia, urine, lines, C. difficile
- Alcohol-associated hepatitis
- GI bleeding
- Drugs: NSAIDs, sedatives, too much diuretic
- HBV flare, ischemia
- No trigger is found in some patients. Repeat the search if he gets worse

EASL ACLF 2023 [PMID: 37364789]; SCCM 2023 [PMID: 37052436]

Why is he hypotensive?

Mechanism	What you do

Surviving Sepsis 2026 [PMID: 41869847]; SCCM 2020 [PMID: 32058387]; EASL ACLF 2023 [PMID: 37364789]

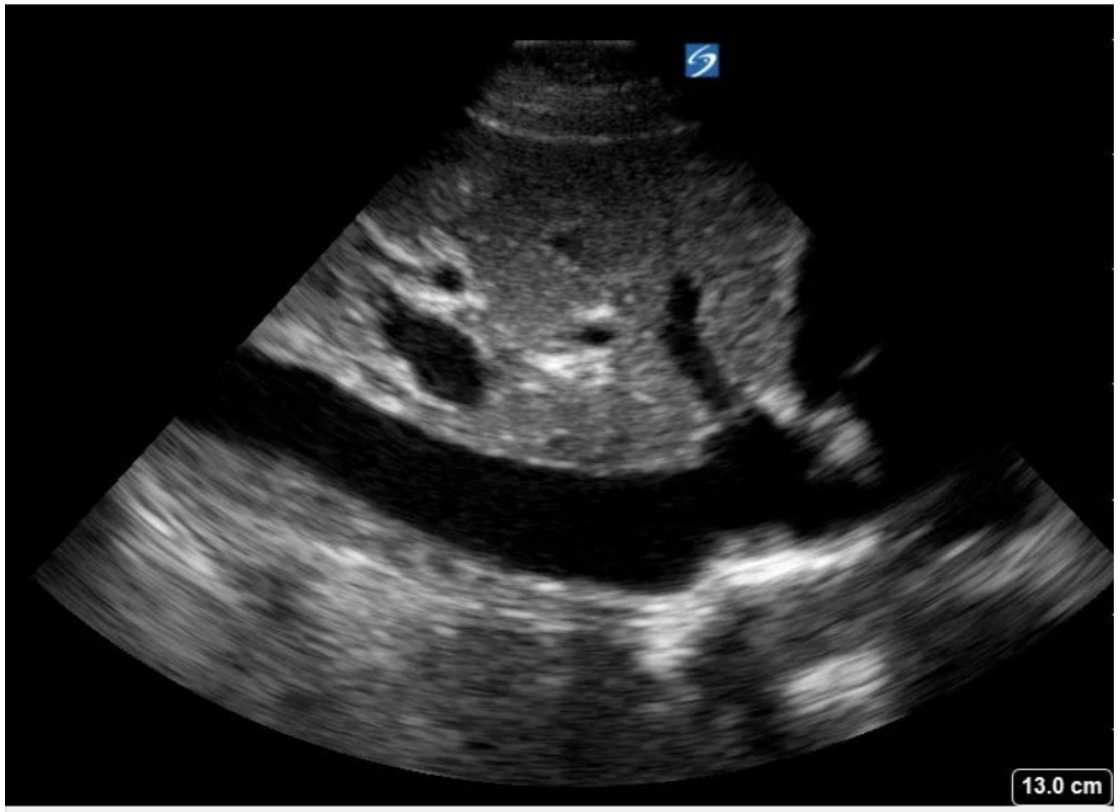
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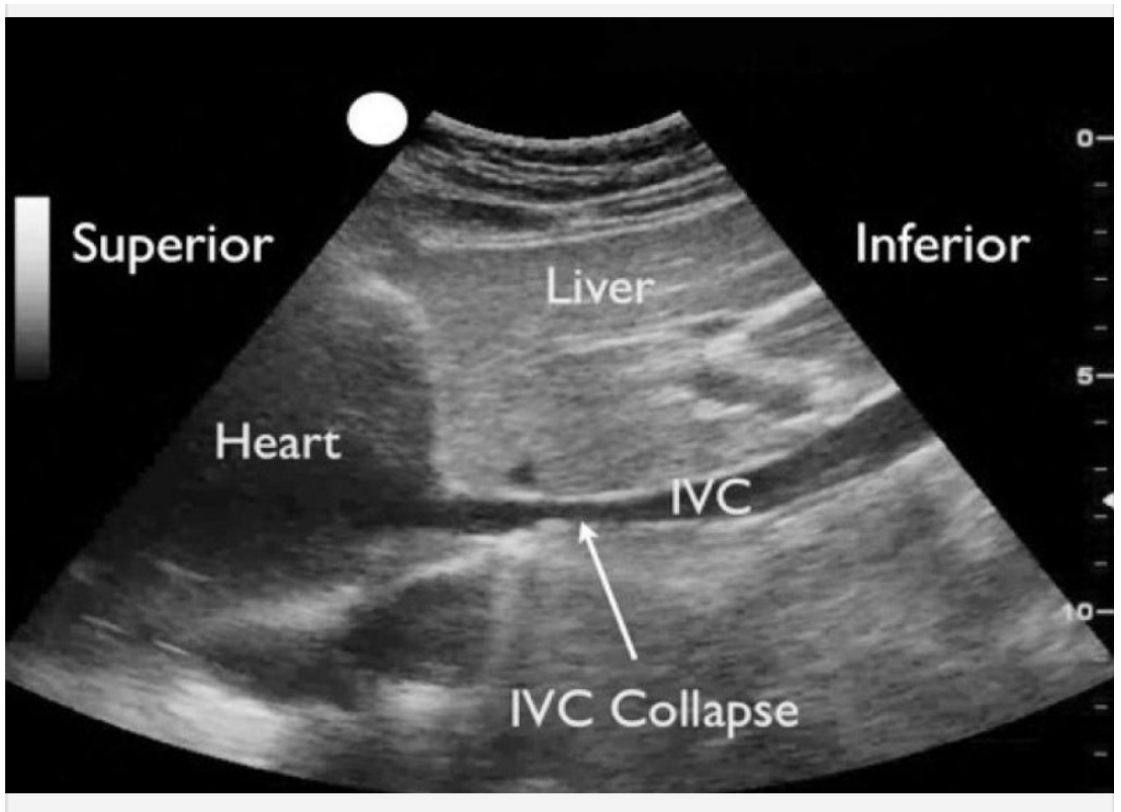
Mechanism	What you do
Sepsis (SBP)	Cultures, antibiotics in the first hour, source control. Cover for prior exposure and local resistance
Splanchnic vasodilation of cirrhosis	Norepinephrine. Start at MAP 65 and adjust to perfusion
Too little volume, or too much	Ultrasound and dynamic measures. Edema does not mean the vessels are full
Hidden bleeding	Hemoglobin, rectal exam, think varices
Relative adrenal insufficiency	Hydrocortisone if shock is refractory
Cirrhotic cardiomyopathy	Echo

Usually more than one of these

Surviving Sepsis 2026 [PMID: 41869847]; SCCM 2020 [PMID: 32058387]; EASL ACLF 2023 [PMID: 37364789]

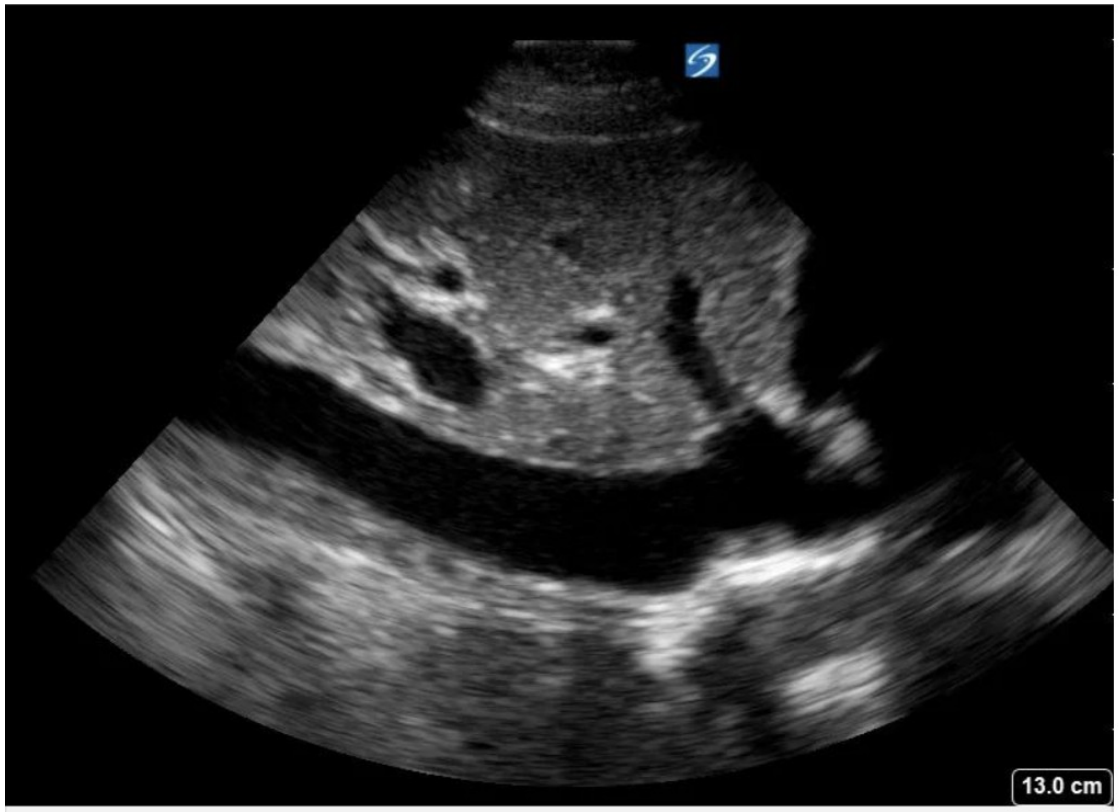
Too little volume, or too much?

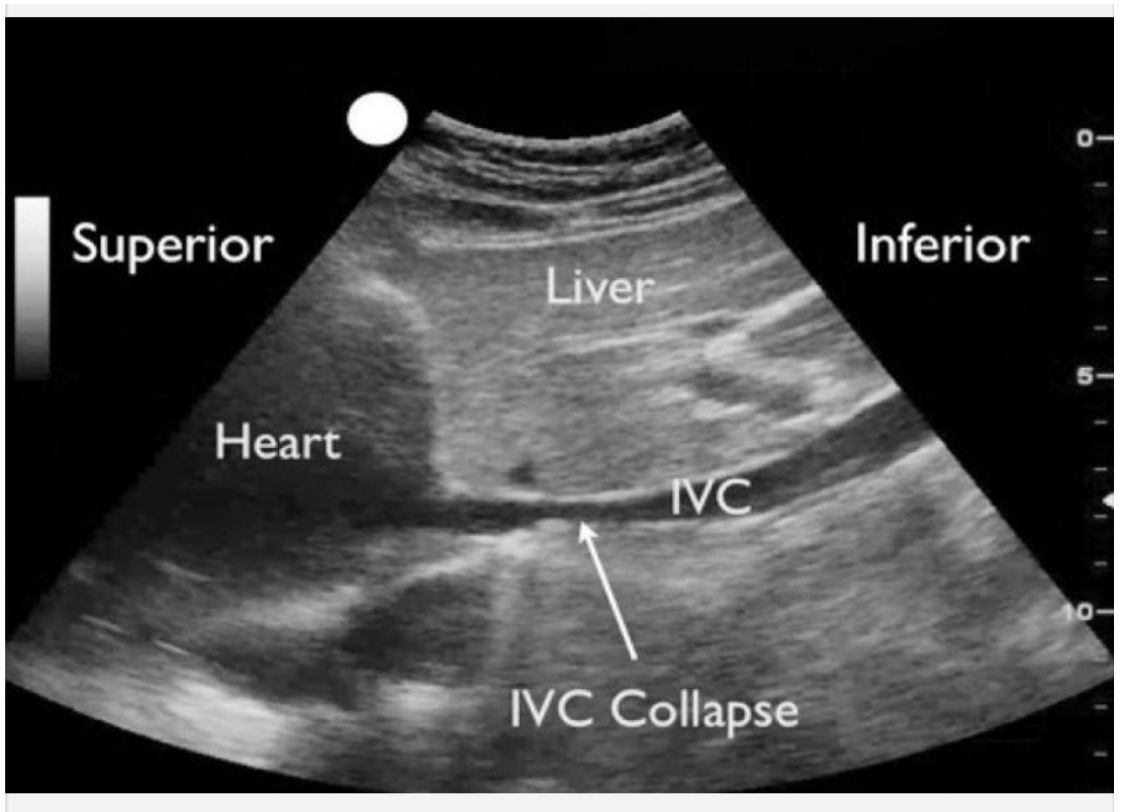




Which one would you give more fluid?

Too little volume, or too much?

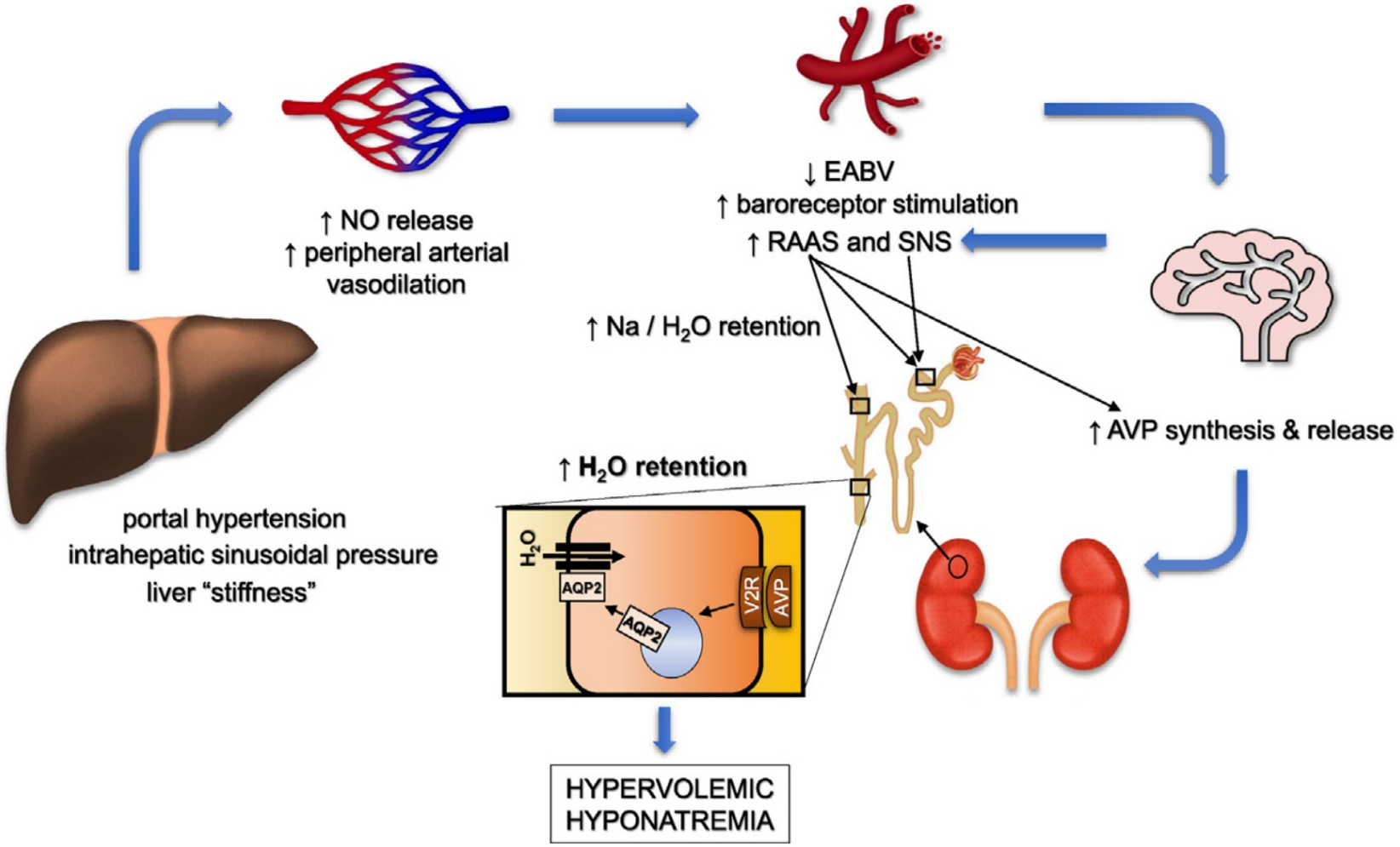




Left: a full, plethoric IVC. Right: an IVC that collapses

Look before you give the next liter

Why is his pressure low to begin with?



Rondon-Berrios H, Velez JCQ. Clin Liver Dis 2022

ALF, decompensation or ACLF?

	Acute liver failure	Acute decompensation	ACLF
Liver before			
What defines it			
What usually kills			
ICU focus			
Transplant			

EASL ALF 2017 [PMID: 28417882]; EASL ACLF 2023 [PMID: 37364789]

ALF, decompensation or ACLF?

	Acute liver failure	Acute decompensation	ACLF
Liver before	No cirrhosis	Cirrhosis	Cirrhosis (or chronic liver disease, depending on definition)
What defines it	INR ≥ 1.5 + encephalopathy	Ascites, bleeding, encephalopathy, jaundice	Decompensation + organ failure
What usually kills	Brain, multiorgan failure, sepsis	Bleeding, infection. Can turn into ACLF	Infection, shock, multiorgan failure
ICU focus	Brain, ammonia, early CRRT	Treat the trigger	Source control, shock, kidney vs lung
Transplant	Urgent listing pathway	Refer when MELD or the complication calls for it	Narrow window; reassess daily

EASL ALF 2017 [PMID: 28417882]; EASL ACLF 2023 [PMID: 37364789]

Count the failing organs

Organ	Failure (EASL-CLIF)
Liver	Bilirubin ≥ 12 mg/dL
Kidney	Creatinine ≥ 2.0 mg/dL or on RRT
Brain	Encephalopathy grade III-IV
Coagulation	INR ≥ 2.5
Circulation	Vasopressors (or terlipressin)
Lungs	PaO2/FiO2 ≤ 200, SpO2/FiO2 ≤ 214, or ventilated (not for encephalopathy)

CLIF-C organ failure score, EASL ACLF 2023 [PMID: 37364789]; CANONIC [PMID: 23474284]

Count the failing organs

Organ	Failure (EASL-CLIF)	Patient B
Liver	Bilirubin ≥ 12 mg/dL	5.8: no
Kidney	Creatinine ≥ 2.0 mg/dL or on RRT	**2.6: yes**
Brain	Encephalopathy grade III-IV	Grade II: no
Coagulation	INR ≥ 2.5	2.1: no
Circulation	Vasopressors (or terlipressin)	**Norepinephrine: yes**
Lungs	PaO2/FiO2 ≤ 200, SpO2/FiO2 ≤ 214, or ventilated (not for encephalopathy)	PaO2/FiO2 211: no

Two organ failures: ACLF grade 2

CLIF-C organ failure score, EASL ACLF 2023 [PMID: 37364789]; CANONIC [PMID: 23474284]

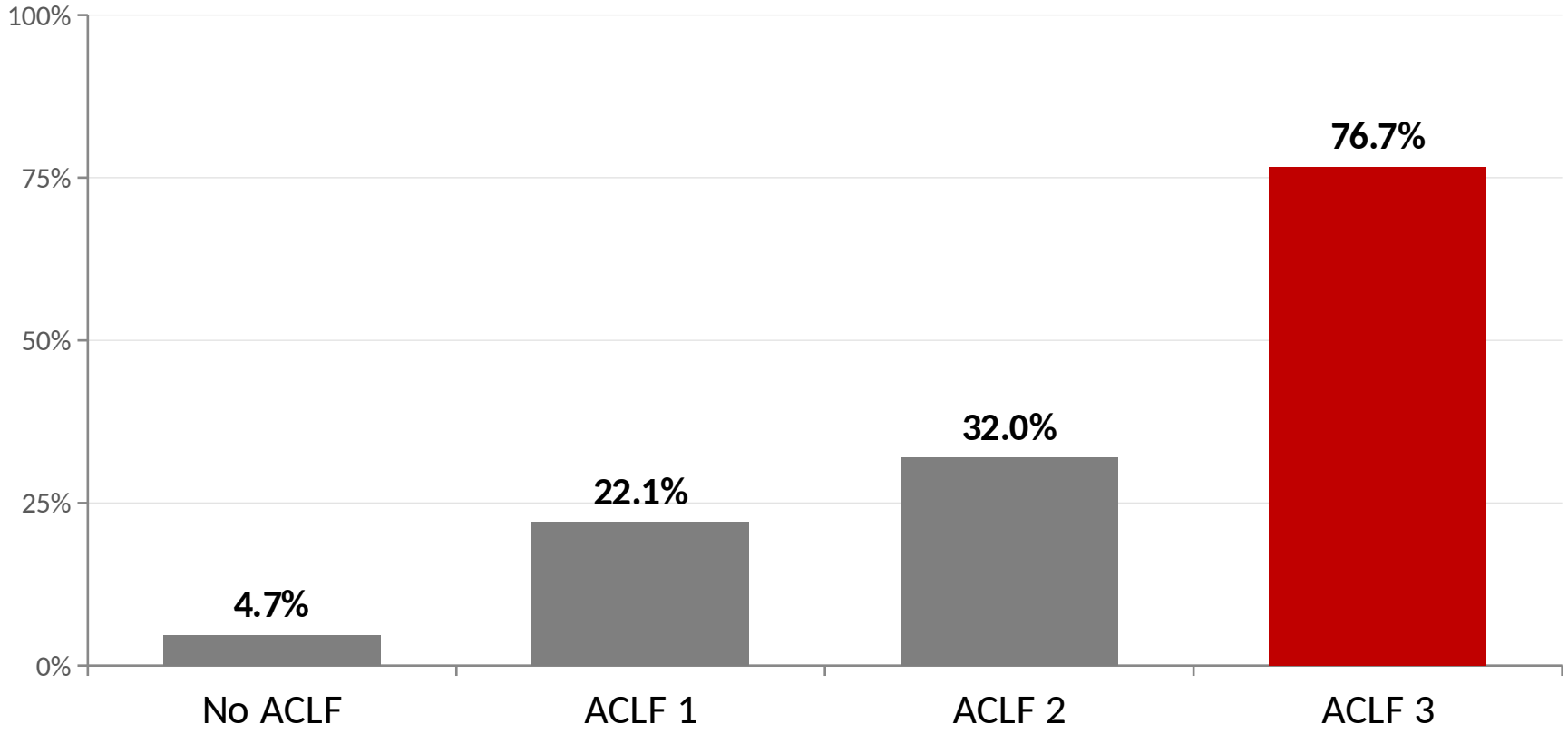
ACLF grades

Grade	What it takes
No ACLF	No organ failure; or one failure of liver, clotting, circulation or lungs, with creatinine < 1.5 and no HE; or brain failure alone, with creatinine < 1.5
ACLF 1	Kidney failure alone; or one failure of liver, clotting, circulation or lungs, with creatinine 1.5 to 1.9 and/or grade I-II HE; or brain failure alone, with creatinine 1.5 to 1.9
ACLF 2	Two organ failures
ACLF 3	Three or more

A patient with "no ACLF" can still be very sick. Use the CLIF-C calculator

Moreau R et al. Gastroenterology 2013 [PMID: 23474284]; EASL ACLF 2023 [PMID: 37364789]

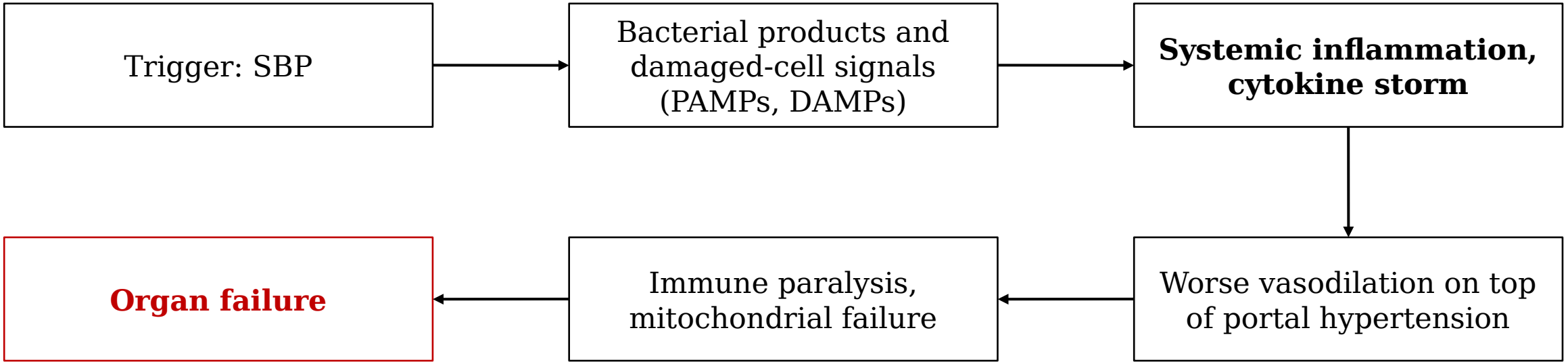
28-day mortality by ACLF grade



CANONIC, 1,343 patients, 2013

Moreau R et al. Gastroenterology 2013 [PMID: 23474284]

Why do the organs fail?



- Bed 7 so far: **kidney** and **circulation**. Lungs and brain are next in line
- Treat the trigger early and support the organs; there is no drug for the inflammation itself

Concept from Arroyo V, Moreau R, Jalan R. N Engl J Med 2020

Does he have HRS?

12 hours later: creatinine 3.0, urine 10 mL/h, still on norepinephrine

HRS? Start terlipressin?

HRS-AKI: what changed in 2024

What many of us learned (ICA 2015)	Now (ADQI-ICA 2024)
No response after 2 days of diuretic withdrawal and albumin 1 g/kg No shock No nephrotoxins, no structural kidney disease	Cirrhosis with ascites AKI by creatinine and/or urine output No improvement within 24 hours of volume resuscitation, if he needs it No albumin challenge if he is already euvolemic or wet No strong alternative explanation as the main cause

Faster to call. But you still have to look for the other causes

Angeli P et al. J Hepatol 2015 (ICA); ADQI-ICA consensus, J Hepatol 2024 [PMID: 38527522]

Does he have HRS?

12 hours later: creatinine 3.0, urine 10 mL/h, still on norepinephrine

HRS-AKI (ADQI-ICA 2024)	**Bed 7, day 1**	
Cirrhosis with ascites		
AKI by creatinine and/or urine output		
No improvement within 24 hours of volume resuscitation, if he needs it		
No strong alternative explanation (shock, nephrotoxins, structural injury)		

Go down the list for him. Which boxes does he tick?

ADQI-ICA consensus, J Hepatol 2024 [PMID: 38527522]

Does he have HRS?

12 hours later: creatinine 3.0, urine 10 mL/h, still on norepinephrine

HRS-AKI (ADQI-ICA 2024)	**Bed 7, day 1**	
Cirrhosis with ascites	Yes	✅
AKI by creatinine and/or urine output	Creatinine 0.9 → 3.0, urine 10 mL/h	✅
No improvement within 24 hours of volume resuscitation, if he needs it	2 L given on arrival. Too early to say	❌
No strong alternative explanation (shock, nephrotoxins, structural injury)	**Septic shock on norepinephrine**	❌

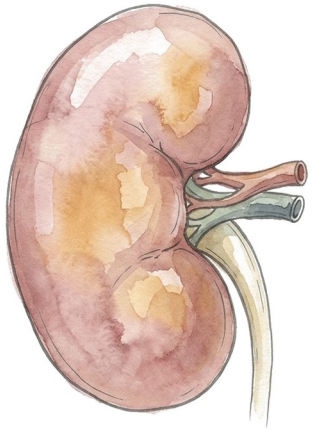
Not HRS yet. Treat the shock first

ADQI-ICA consensus, J Hepatol 2024 [PMID: 38527522]

Kidney failure in cirrhosis is usually a mixture

Possible contributors

- Low effective volume
- Tubular injury: sepsis, contrast, drugs
- HRS physiology
- Tense ascites (abdominal compartment)
- Congestion

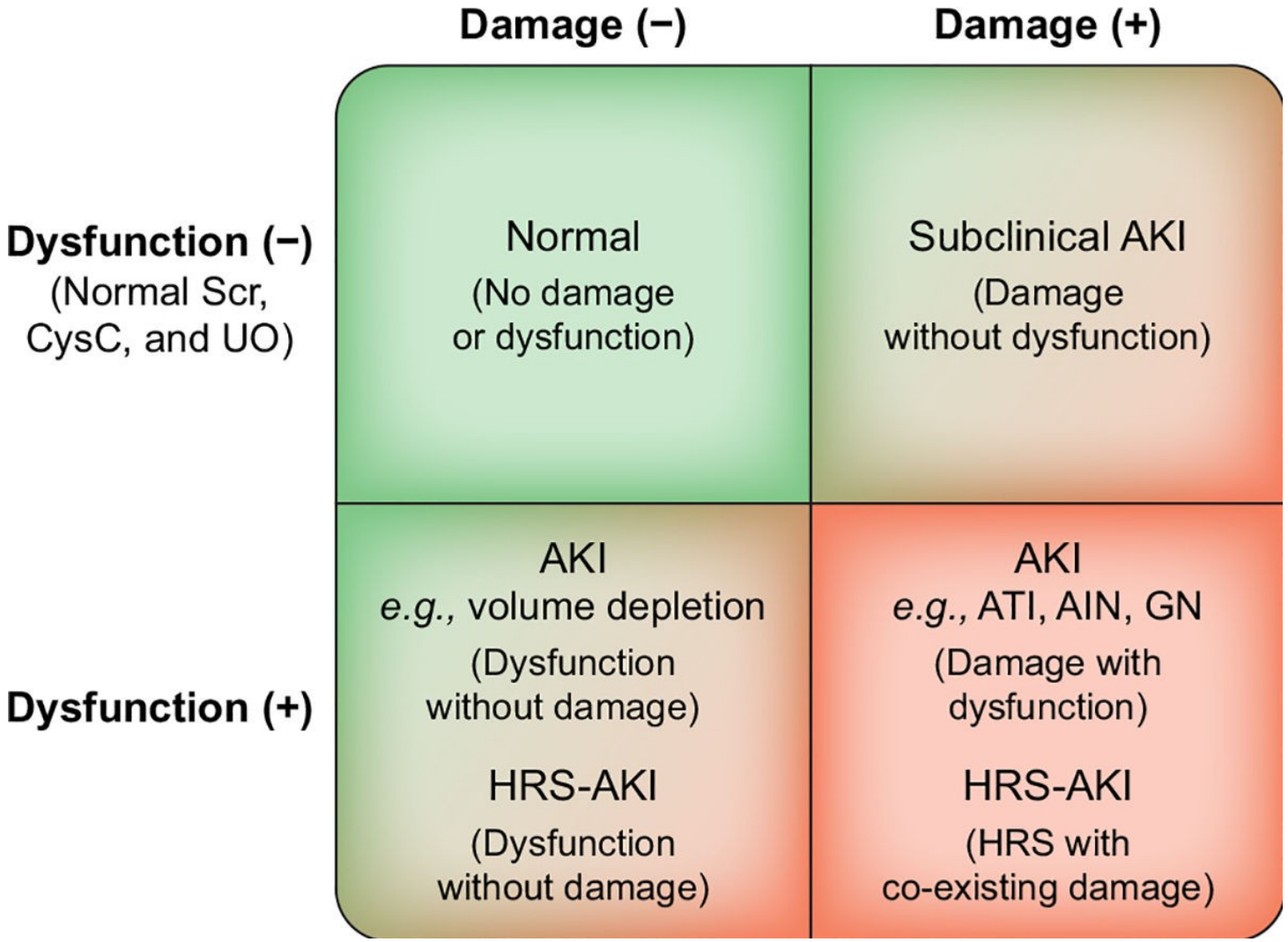


Look

- Urinalysis and sediment
- Drug list
- Ultrasound: kidneys, bladder, IVC, heart, lungs
- Bladder pressure if the belly is tight
- A low urine sodium is **not** proof of HRS

ADQI-ICA consensus 2024 [PMID: 38527522]

Function and damage: four kinds of AKI



Nadim MK et al. J Hepatol 2024;81:163-83 [PMID: 38527522], Fig. 4 (CC BY 2.0, adapted from ADQI)

Day 1: more confused

- More disoriented, pulling at his lines, asterixis. Awake, follows commands
- The nurse sent an ammonia: **88 µmol/L**

More lactulose? Repeat the ammonia in 6 hours?

Day 1: more confused

- More disoriented, pulling at his lines, asterixis. Awake, follows commands
- The nurse sent an ammonia: **88 µmol/L**

Look for the reason first. The ammonia doesn't tell you

- Glucose, sodium, oxygen and CO2
- Is the infection controlled? Cultures, source, antibiotic coverage
- Bleeding, sedatives and opioids, withdrawal, seizures, a new focal deficit (CT head)
- Lactulose for overt HE: 2 to 3 soft stools a day. By tube if he can't swallow safely; watch for dehydration and high sodium
- Rifaximin is an add-on, mainly to prevent the next episode
- Grade III-IV and a poor airway: intubate
- No repeat ammonia

EASL HE guideline 2022 [PMID: 35724930]; SCCM 2023 [PMID: 37052436]

Day 2: worse kidney, worse lungs

	Admission	Day 2
Albumin received		110 g (SBP dose, 1.5 g/kg at 73 kg) + 2 L crystalloid
Breathing	SpO2 92%, Venturi 35%	**SpO2 86% on Venturi 50%, RR 32, crackles**
Ultrasound		Diffuse B-lines, IVC full
Creatinine (mg/dL)	2.6	3.1
Urine output	15 mL/h	10 mL/h
Norepinephrine (mcg/kg/min)	0.15	0.22

More albumin? Terlipressin? Something else?

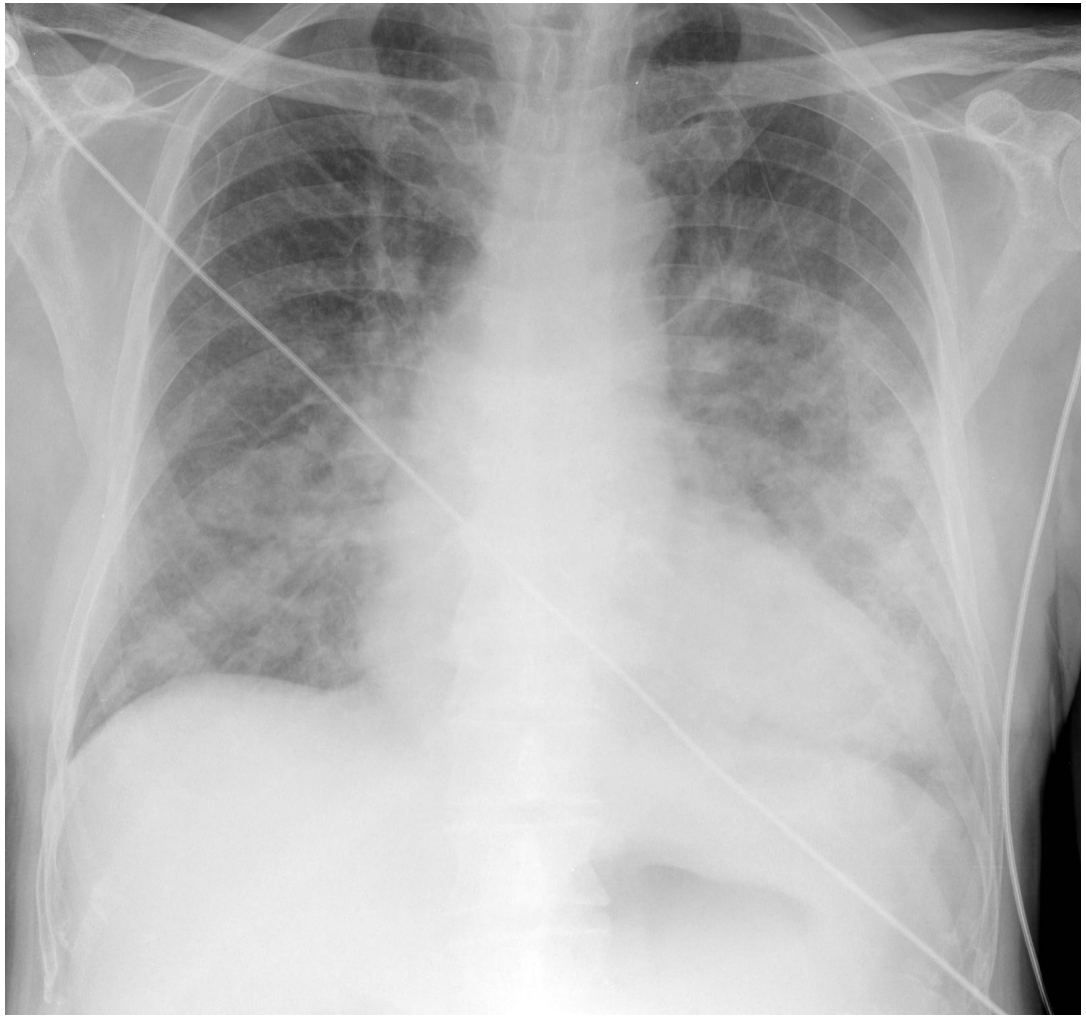
Day 2: worse kidney, worse lungs

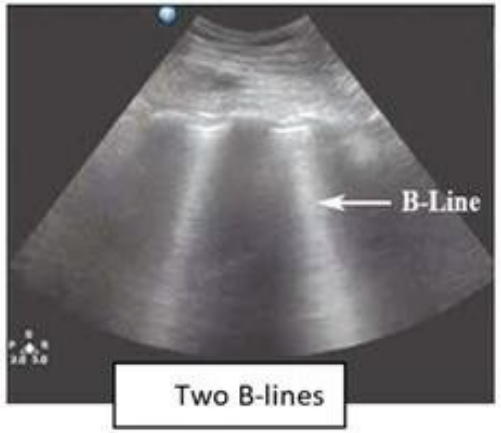
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Creatinine (mg/dL)	2.6	3.1
Urine output	15 mL/h	10 mL/h
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First, what are the lungs telling you?

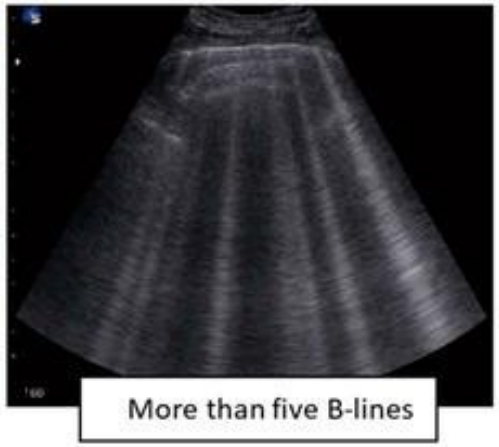
Now on high-flow, FiO2 0.6, SpO2 90%: SpO2/FiO2 150. That's a third organ failure

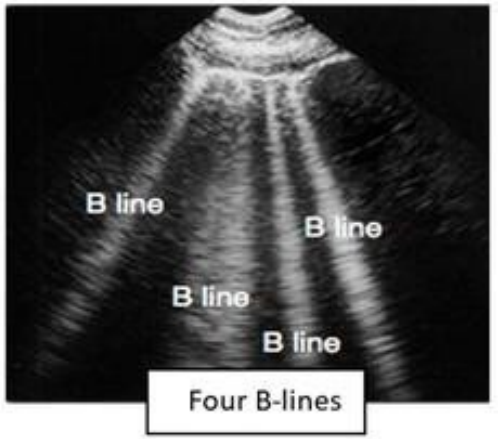
Day 2: what do his lungs look like?











Images for teaching, not this patient

Will fixing the kidney hurt the lungs?

Terlipressin, more albumin, dialysis: what would you do for him today?

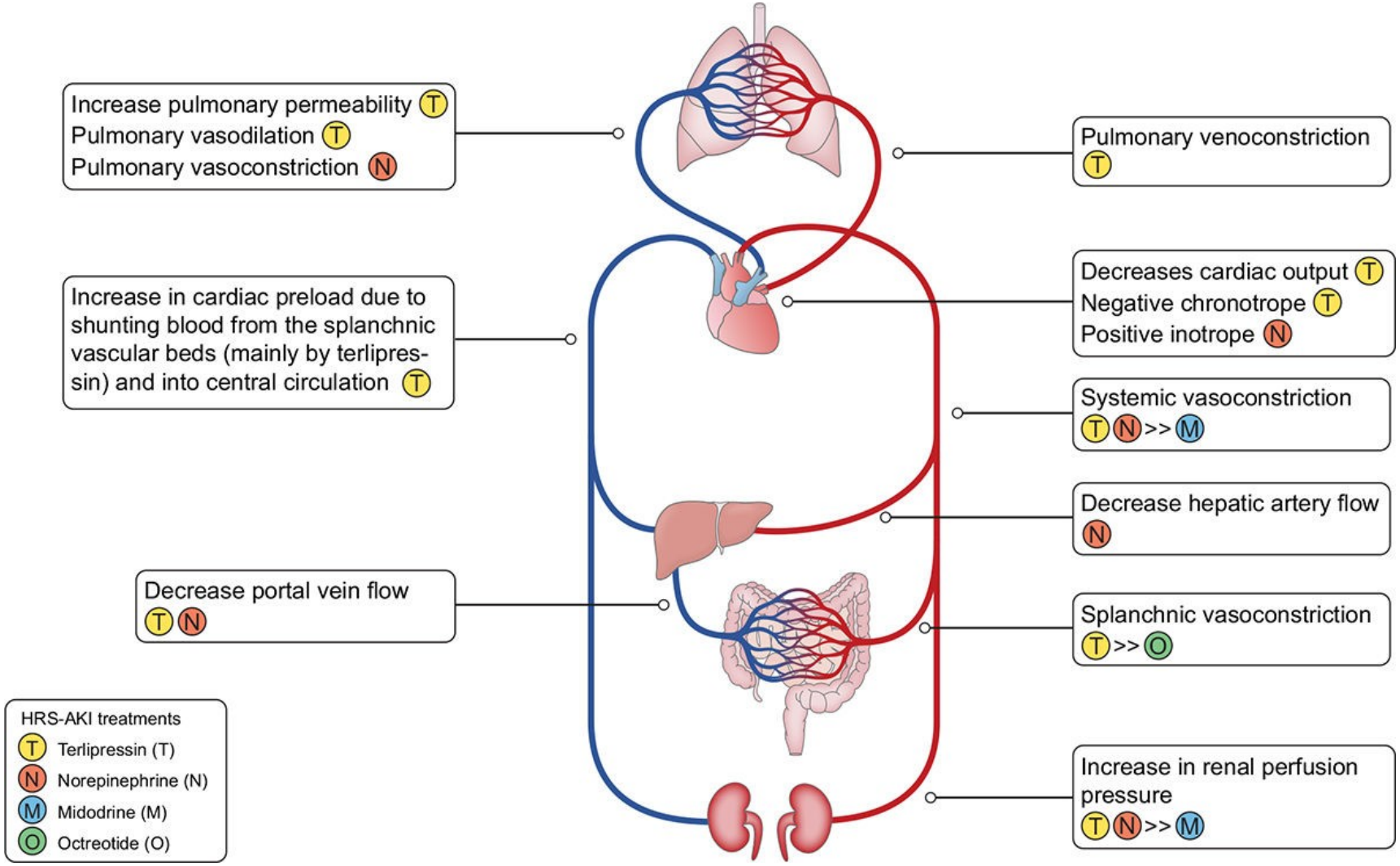
Will fixing the kidney hurt the lungs?

- **Terlipressin: not for him today**
  - Label: contraindicated with hypoxia or worsening respiratory symptoms; don't start if SpO2 < 90%, stop if it falls below 90%
  - Label: avoid in ACLF grade 3; volume overload raises the risk of respiratory failure
- Stop the albumin while he is wet
- He is already on a vasoconstrictor. Norepinephrine, titrated to perfusion
- RRT if he stays overloaded. For HRS, RRT is a bridge to transplant: if he isn't a candidate, talk about goals before you start
- Midodrine and octreotide? They work poorly, and not in someone on IV pressors

Check oxygen and volume before any kidney treatment

TERLIVAZ prescribing information, revised 6/2026; ADQI-ICA 2024 [PMID: 38527522]

What each HRS drug does to the heart and lungs



Nadim MK et al. J Hepatol 2024;81:163-83 [PMID: 38527522], Fig. 5 (CC BY 2.0, adapted from ADQI)

Does HRS reversal mean survival?

The NEW ENGLAND JOURNAL of MEDICINE

ORIGINAL ARTICLE

Terlipressin plus Albumin for the Treatment of Type 1 Hepatorenal Syndrome

F. Wong, S.C. Pappas, M.P. Curry, K.R. Reddy, R.A. Rubin, M.K. Porayko, S.A. Gonzalez, K. Mumtaz, N. Lim, D.A. Simonetto, P. Sharma, A.J. Sanyal, M.J. Mayo, R.T. Frederick, S. Escalante, and K. Jamil, for the CONFIRM Study Investigators*

What do you think they found?

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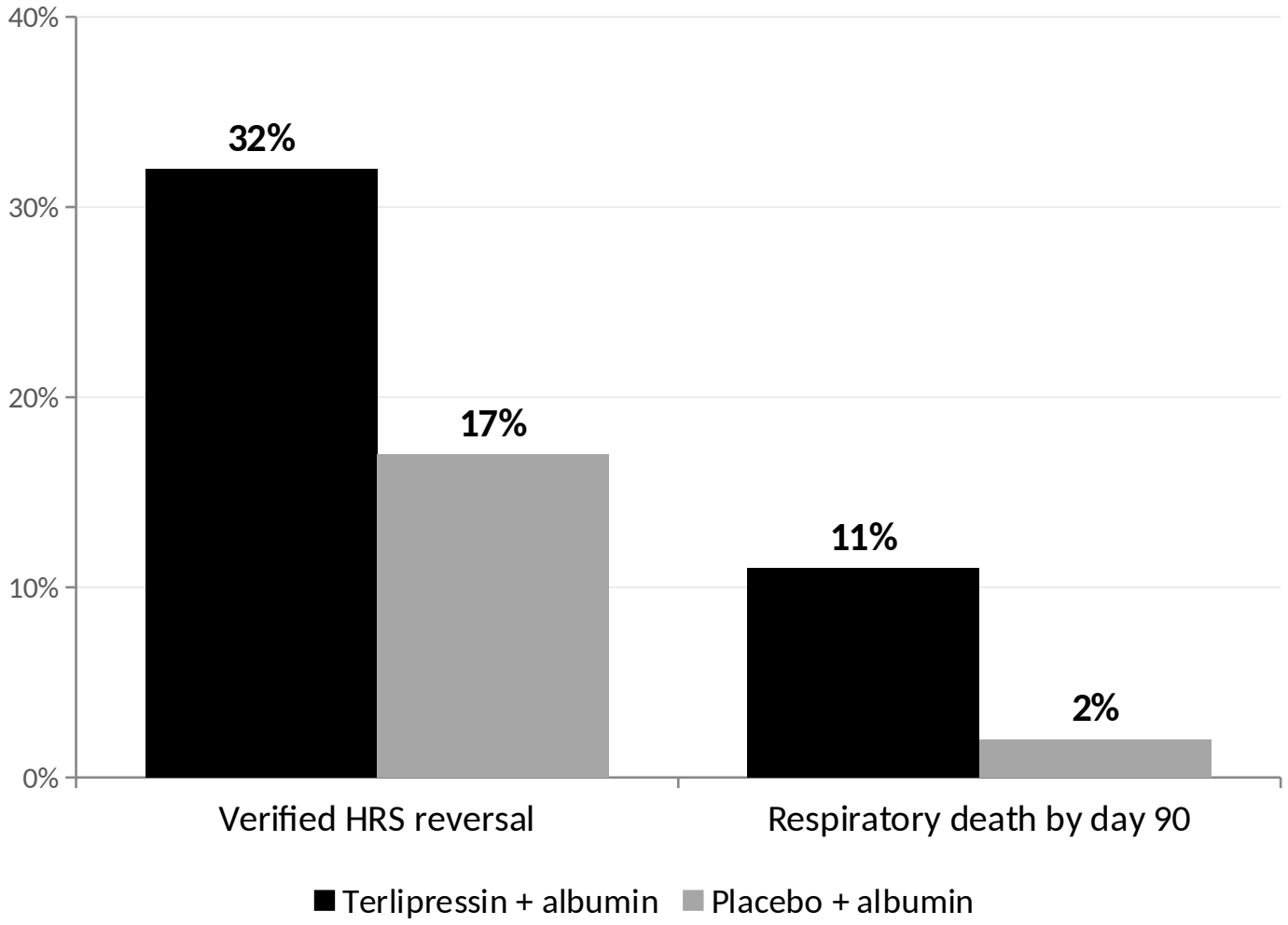
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CONCLUSIONS

In this trial involving adults with cirrhosis and HRS-1, terlipressin was more effective than placebo in improving renal function but was associated with serious adverse events, including respiratory failure. (Funded by Mallinckrodt Pharmaceuticals; CONFIRM ClinicalTrials.gov number, NCT02770716.)

Wong F et al. N Engl J Med 2021;384:818-28 [PMID: 33657294]

Does HRS reversal mean survival?



CONFIRM: type 1 HRS, 300 patients

- More kidneys recovered
- **More patients died of respiratory failure**
- All-cause 90-day mortality was not different
- Respiratory failure as an adverse event: about 14% vs 5% (label)

Wong F et al. NEJM 2021 [PMID: 33657294]; TERLIVAZ prescribing information, revised 6/2026

Should we treat the albumin number?

The NEW ENGLAND JOURNAL of MEDICINE

ORIGINAL ARTICLE

A Randomized Trial of Albumin Infusions in Hospitalized Patients with Cirrhosis

Louise China, Ph.D., Nick Freemantle, Ph.D., Ewan Forrest, M.D., Yiannis Kallis, Ph.D., Stephen D. Ryder, D.M., Gavin Wright, Ph.D., Andrew J. Portal, M.D., Natalia Becares Salles, Ph.D., Derek W. Gilroy, Ph.D., and Alastair O'Brien, Ph.D., for the ATTIRE Trial Investigators*

What do you think they found?

China L et al. N Engl J Med 2021;384:808-17 [PMID: 33657293]

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CONCLUSIONS

In patients hospitalized with decompensated cirrhosis, albumin infusions to increase the albumin level to a target of 30 g per liter or more was not more beneficial than the current standard care in the United Kingdom.

China L et al. N Engl J Med 2021;384:808-17 [PMID: 33657293]

Should we treat the albumin number?

Study	ATTIRE: daily albumin to keep serum albumin ≥ 30 g/L vs standard care, hospitalized decompensated cirrhosis, 777 patients
Main finding	Composite of new infection, kidney dysfunction or death: 29.7% vs 30.2%
The catch	**More serious adverse events with albumin, including fluid overload and pulmonary edema**

When albumin is indicated

- SBP: 1.5 g/kg on day 1, 1 g/kg on day 3
- Paracentesis over 5 L: 6 to 8 g per liter removed
- HRS-AKI: with a vasoconstrictor, dosed to volume status

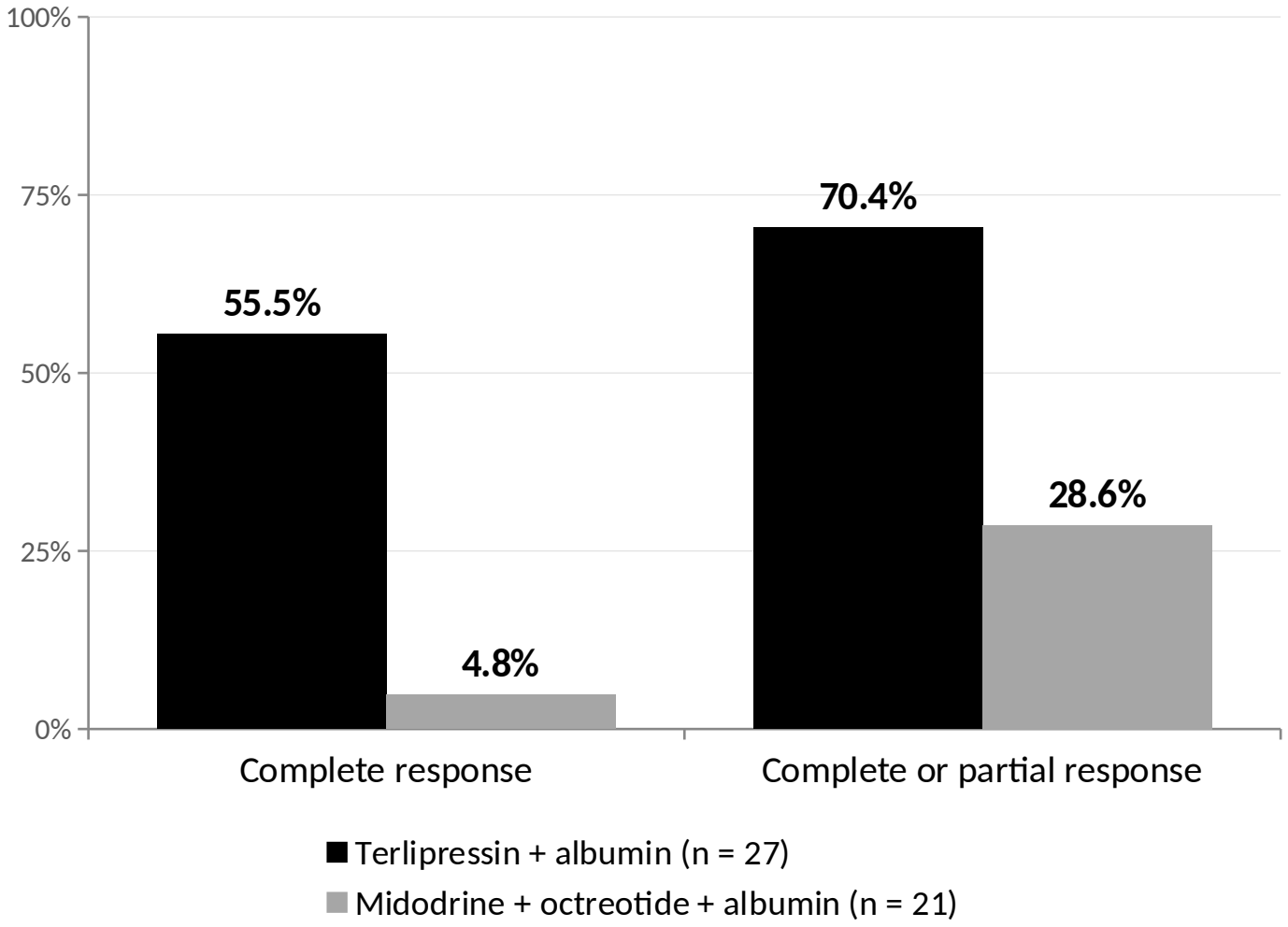
China L et al. NEJM 2021 [PMID: 33657293]; ADQI-ICA 2024 [PMID: 38527522]

Midodrine and octreotide for HRS-AKI?

Head to head: terlipressin vs midodrine plus octreotide, both with albumin, type 1 HRS

How different can two vasoconstrictors be?

Midodrine and octreotide for HRS-AKI?



Randomized, 8 Italian centers, type 1 HRS. 49 randomized (27 vs 22), 48 analyzed

- Stopped early at the interim analysis because the difference was already clear
- Blood pressure rose more with terlipressin
- 3-month survival 59% vs 43%; too small to tell
- **Use it only if terlipressin and norepinephrine are not options**

Cavallin M et al. Hepatology 2015;62:567-74 [PMID: 25644760]; ADQI-ICA 2024 [PMID: 38527522]

What changed since yesterday?

	Day 1	Day 2	Day 3	Day 5
Norepinephrine (mcg/kg/min)	0.15	0.22	0.12	Off
Oxygen, SpO2	Venturi 0.35, 92%	HFNC 0.6, 90%	HFNC 0.4, 95%	2 L, 97%
Lung failure?	No (P/F 211)	Yes (S/F 150)	No (S/F 238)	No
Kidney	Cr 2.6	Cr 3.1	CRRT for fluid	On CRRT
Encephalopathy	Grade II	Grade II	Grade I	Grade I
Bilirubin / INR	5.8 / 2.1	6.4 / 2.3	6.1 / 2.2	5.2 / 2.0
ACLF grade	2	**3**	2	?

Better or worse? What do you tell his family?

Made-up numbers for teaching. CLIF-C ACLF score [PMID: 24950482]; EASL ACLF 2023 [PMID: 37364789]

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Bilirubin / INR	5.8 / 2.1	6.4 / 2.3	6.1 / 2.2	5.2 / 2.0
ACLF grade	2	**3**	2	**1**

- The grade at days 3 to 7 predicts outcome better than the grade at admission
- Persistent ACLF 3 at days 3 to 7, especially with four or more organ failures, has a very high mortality without transplant
- If he is not a transplant candidate, discuss goals of care with the family and involve palliative care

Made-up numbers for teaching. CLIF-C ACLF score [PMID: 24950482]; EASL ACLF 2023 [PMID: 37364789]

Who gets a liver?

Transplant in ACLF



Too sick for a transplant, or sick enough to need one?

Ventilator, pressors, CRRT, ACLF 3. Is he too sick?

Too sick for a transplant, or sick enough to need one?

- A ventilator, pressors or dialysis do not by themselves exclude transplant
- Selected patients with ACLF grade 3 do well after transplant. The window is short

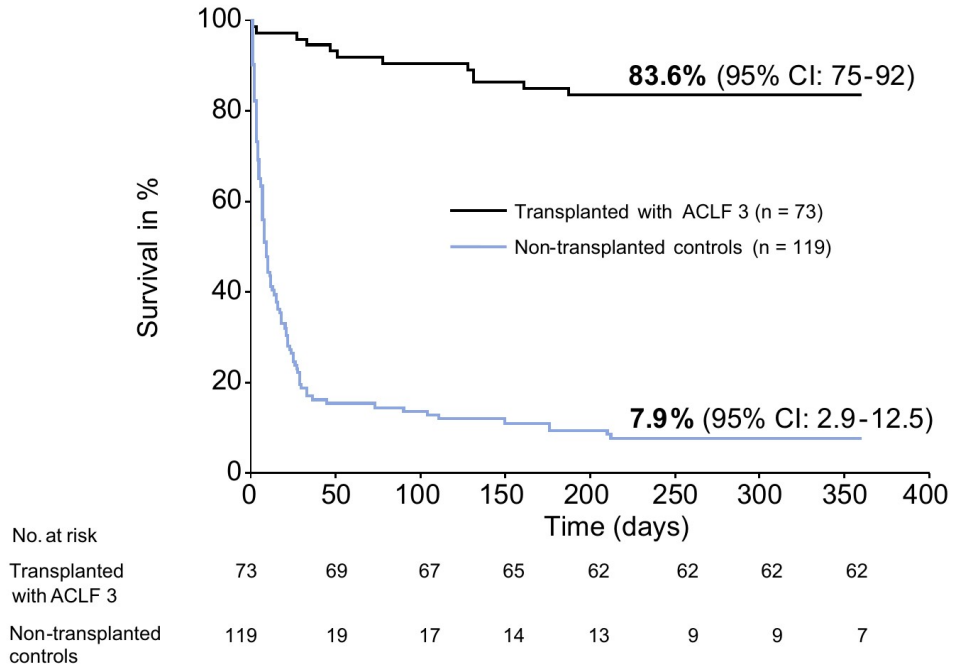
What can close the window

- Uncontrolled infection
- Rising vasopressor needs, severe respiratory failure
- Irreversible brain injury
- Frailty and comorbidity
- Alcohol: centers now evaluate individually rather than by a fixed abstinence rule

Research Article

EASL JOURNAL OF HEPATOLOGY

Liver transplantation in the most severely ill cirrhotic patients: A multicenter study in acute-on-chronic liver failure grade 3



Selected patients, not randomized

Artru F et al. J Hepatol 2017 [PMID: 28645736]; Lancet Gastroenterol Hepatol 2024 [PMID: 38309288]; AASLD/AST 2026 [PMID: 41405234]

Calling transplant hepatology

Call first. Gather this while you wait for the call back

Patient A (ALF)	Patient B (ACLF)
Cause and time course (days from jaundice to encephalopathy) Encephalopathy grade, pupils, sedation Airway and ventilator settings Ammonia, sodium, CRRT INR, lactate, pH, glucose trends Pressors, infection Social support, intent of ingestion	Cause of cirrhosis, recent alcohol use Precipitant and source control Organ failures today and yesterday Pressor dose, oxygen, RRT MELD, CLIF-C scores Frailty, comorbidities Prior transplant evaluation, support

ALF and ACLF: what changes at the bedside

Both have a sick liver. Do they have the same disease?

	Acute liver failure	ACLF
Ammonia		
Lactulose		
Sodium		
Antibiotics		
INR		

ACG ALF 2023 [PMID: 37377263]; SCCM 2023 [PMID: 37052436]; update 2022 [PMID: 36001996]

ALF and ACLF: what changes at the bedside

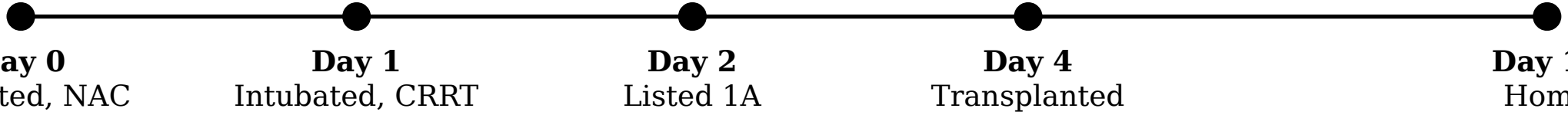
Both have a sick liver. Do they have the same disease?

	Acute liver failure	ACLF
Ammonia	With the HE grade and the trend: brain risk, and when to start CRRT. No level rules out ICH	Doesn't diagnose HE. Don't trend it to dose lactulose
Lactulose	No proven benefit; bowel distension can get in the way of surgery	Yes for overt HE, plus treat the trigger. Rifaximin as an add-on. Avoid dehydration
Sodium	Grade III-IV or high ammonia: center protocol, often 145 to 155, matched to the CRRT fluid	Often chronically low: correct slowly. In high-risk patients, no more than 8 mmol/L in 24 hours
Antibiotics	No routine prophylaxis. Surveillance cultures, low threshold to treat	Treat infection fast; prophylaxis with GI bleeding
INR	Tracks severity, poor at predicting bleeding. No plasma just for the number	Same. Judge the procedure and the patient

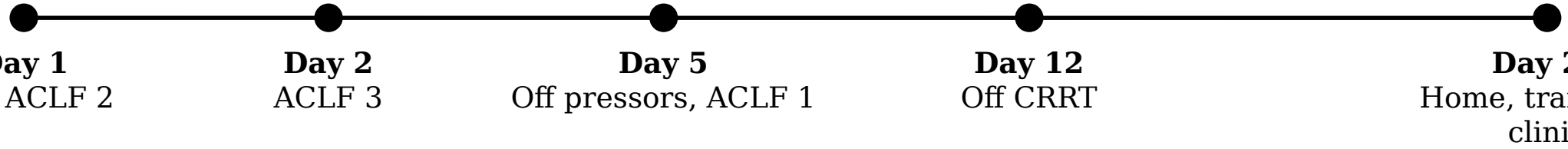
ACG ALF 2023 [PMID: 37377263]; SCCM 2023 [PMID: 37052436]; update 2022 [PMID: 36001996]

What happened to them

Bed 4



Bed 7



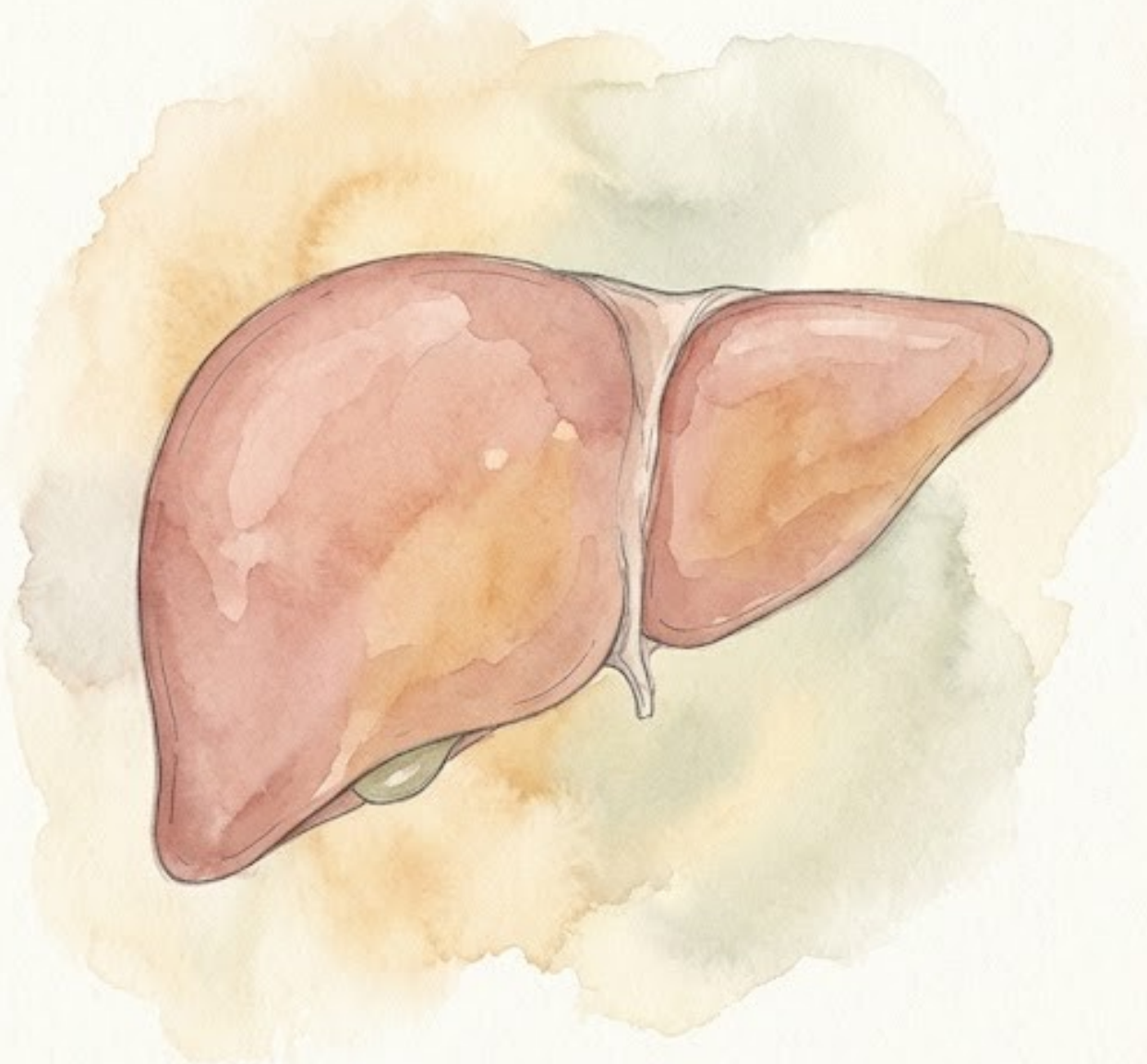
Made-up outcomes for teaching

Back to our patients

- Which one needed transplant hepatology before the tests came back?
- Why did a falling ALT make you more worried?
- When is CRRT right before the creatinine is high?
- Why isn't lactulose for ACLF evidence for ALF?
- Why were albumin and terlipressin the wrong next step for bed 7?
- What would make you change yesterday's plan?

Some tricks of the trade

- High AST and ALT alone are not liver failure. Get an INR and examine the patient
- Stabilize, treat the cause and call transplant hepatology in the same hour
- ALF: **protect the brain**. Intubate at grade III, keep the CO2 normal, follow your center's sodium plan, and start CRRT for rising ammonia
- ACLF: count the organs, find the trigger
- ACLF: check oxygenation and volume status before albumin or terlipressin
- Reassess organ failures daily. The trend matters more than the admission score



Questions?

Appendix: West Haven grades

Grade	What you see
I	Short attention, mild confusion, reversed sleep, subtle personality change
II	Lethargy, disorientation to time, obvious personality change, asterixis
III	Somnolent but rousable to voice or pain, grossly disoriented, bizarre behavior
IV	Coma

Appendix: King's College criteria

Acetaminophen	Not acetaminophen
Arterial pH < 7.30 after fluids *or all three within 24 hours:* INR > 6.5 Creatinine > 3.4 mg/dL Grade III-IV encephalopathy *(Modified: lactate > 3.5 at 4 h or > 3.0 at 12 h after fluids)*	**INR > 6.5** *or any three of:* Age < 10 or > 40 Drug reaction, or indeterminate cause Jaundice to encephalopathy > 7 days INR > 3.5 Bilirubin > 17.5 mg/dL

Original O'Grady criteria; the lactate line is the later modification

O'Grady 1989 criteria, as summarized in EASL 2017 [PMID: 28417882] and update 2022 [PMID: 36001996]

Appendix: arterial ammonia studies in ALF

Study (arterial ammonia)	Patients	What they found	The catch
Clemmesen 1999, Copenhagen	44 ALF	**14 herniated. Mean ammonia 230 vs 118 µmol/L; every herniation was above 146**	Small, one center
Bhatia 2006, Delhi	80 ALF	Admission ammonia ≥ 124 predicted death (sensitivity 79%, specificity 76%)	One center; the endpoint was death, not ICH
Bernal 2007, King's	165 ALF, grade III-IV	> 100 predicted severe HE (70% accuracy). **> 200: ICH in 55%**	200 missed most of the ICH cases
Kumar 2012, Delhi	295 ALF	**≥ 122 for 3 days: survival 23% vs 72%**, cerebral edema 71% vs 37%	One center, but the trend beat a single value

Above 150 to 200 and not falling: worry about the brain

ICH can occur below 200. Interpret the level with the HE grade and the trend

Clemmesen 1999 [PMID: 10051463]; Bhatia 2006 [PMID: 16024550]; Bernal 2007 [PMID: 17685471]; Kumar 2012 [PMID: 22521861]

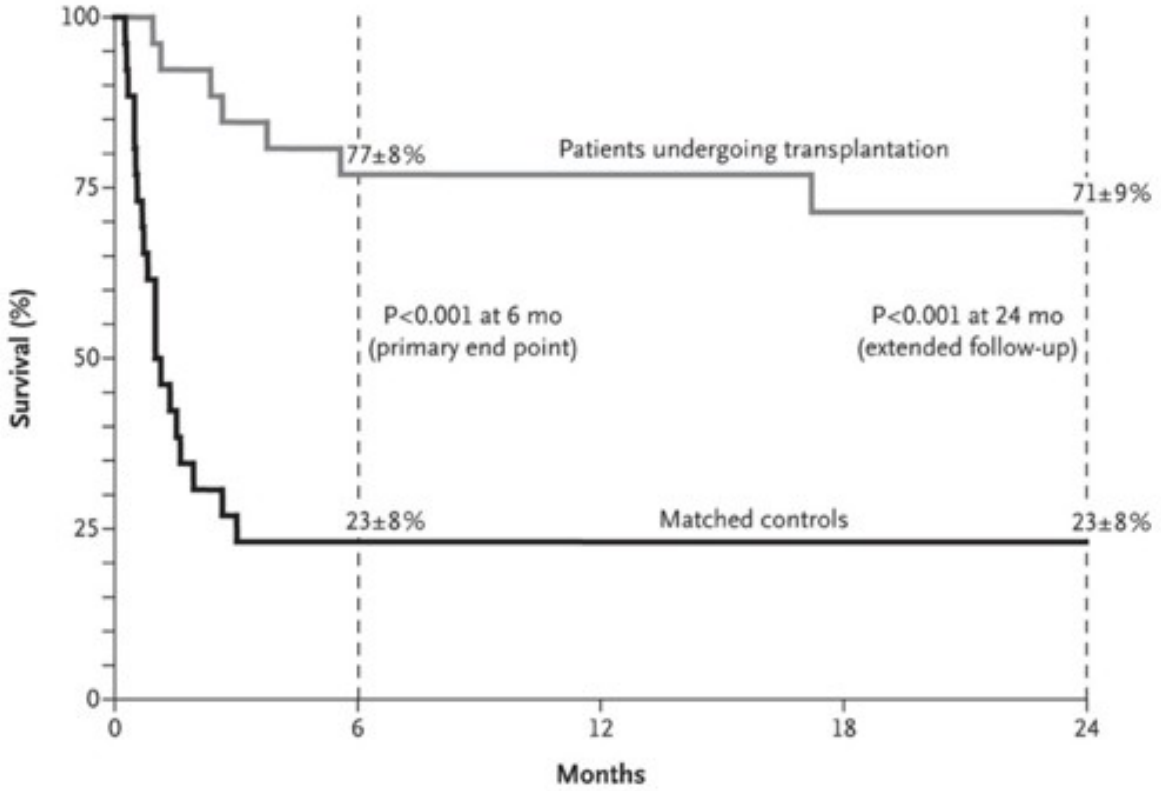
Appendix: which ACLF?

	EASL-CLIF	NACSELD	APASL
Who	Cirrhosis with acute decompensation	Cirrhosis, hospitalized	Chronic liver disease, with or without cirrhosis
Defined by	Organ failures (CLIF-C)	Two or more extrahepatic failures: shock, grade III-IV HE, dialysis, ventilation	Jaundice (bilirubin ≥ 5) and INR ≥ 1.5, then ascites or HE within 4 weeks
Picks up	Broader, earlier	Sicker patients	A primary liver insult

They find different patients. Say which one you're using

Kyoto consensus 2025 [PMID: 39961976]; ACG ACLF 2022 [PMID: 35006099]; AASLD 2024 [PMID: 37939273]

Appendix: early transplant for alcohol-associated hepatitis



No. at Risk					
Patients undergoing transplantation	26	20	15	14	13
Matched controls	26	6	6	5	4

- 26 patients with a first episode of severe alcohol-associated hepatitis, not responding to steroids
- Highly selected: first liver-related event, close family support, no severe other illness, agreed to abstinence
- 6-month survival **77% vs 23%**
- Recent drinking alone is not a reason to skip the call

Mathurin P et al. N Engl J Med 2011 [PMID: 22070476]

Appendix: sources (1)

- SCCM ALF/ACLF ICU guidelines, part 1. Crit Care Med 2020 [PMID: 32058387]
- SCCM ALF/ACLF ICU guidelines, part 2. Crit Care Med 2023 [PMID: 37052436]
- EASL acute liver failure guideline. J Hepatol 2017 [PMID: 28417882]
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- EASL ACLF guideline. J Hepatol 2023 [PMID: 37364789]
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- EASL hepatic encephalopathy guideline. J Hepatol 2022 [PMID: 35724930]
- US/Canada acetaminophen poisoning consensus. JAMA Netw Open 2023 [PMID: 37552484]

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- Mathurin P et al. Early transplant for alcoholic hepatitis. NEJM 2011 [PMID: 22070476]
- China L et al. ATTIRE. NEJM 2021 [PMID: 33657293]
- Liver transplantation for ACLF. Lancet Gastroenterol Hepatol 2024 [PMID: 38309288]
- AASLD/AST candidate evaluation guideline. Hepatology 2026 [PMID: 41405234]
- Kyoto consensus on ACLF. Hepatol Int 2025 [PMID: 39961976]
- TERLIVAZ (terlipressin) prescribing information, revised 6/2026